

The Dollar's Safe-Haven Status Across Two Shocks: Evidence from Liberation Day and the Strait of Hormuz Crisis

Bachelor's Thesis

Department of Economics

University of Mannheim

submitted to:

Dr. Husnu C. Dalgic

submitted by:

Leon Weigele

Matriculation Number: 1957464

Degree Program: Bachelor of Science in Economics (B.Sc.)

Address: Stamitzstraße 5, 68167 Mannheim

Phone Number: +49 1756208327

Email: leon.weigele@students.uni-mannheim.de

Mannheim, September 15, 2026

Contents

1	Introduction	1
2	Literature Review	4
2.1	Currency Risk Factors	4
2.2	Dollar Dominance and Its Erosion	5
2.3	Safe-Haven Currencies and Gold	6
2.4	Geopolitical Risk	7
2.5	Tariff Shocks and Exchange Rates	7
2.6	Commodity Currencies and Oil	9
3	Conceptual Framework	10
3.1	The Dollar as a Safe Haven	10
3.2	The Limits of Dollar Safety	11
3.3	Design of the Comparison	12
3.4	Hypotheses	13
4	Background	14
4.1	Liberation Day	14
4.2	Twelve-Day War and Hormuz Crisis	15
4.3	Russian Invasion of Ukraine	16
4.4	Overview of the Events	16
5	Data and Methodology	19
5.1	Data	19
5.1.1	Exchange Rates and Financial Variables	19
5.1.2	Geopolitical and Policy Uncertainty Indices	20
5.1.3	Portfolio Construction	20
5.2	Event Study Methodology	22
5.3	Regression Specification	24
6	Empirical Results	26
6.1	Dollar Response to the Tariff Shock	26
6.2	Dollar Response to the Military Shocks	29
6.3	Cross-Event Comparison	33
6.4	Oil, Gold and Equities	35
6.5	Robustness	37

Contents

6.6	Regression Results	40
7	Discussion	42
7.1	Findings in Light of the Literature	42
7.2	The Dollar's Safe Haven Is Conditional on Shock Type	43
7.3	Gold and the Oil Channel	44
7.4	Policy Implications	45
7.5	Limitations	45
8	Conclusion	46
9	References	48
10	Data Sources	53
	Appendix	55
1	Currency Universe and Classification	55
2	Robustness	57

1 Introduction

Within a single year, two major shocks moved global commodity and financial markets. On the 2nd of April 2025, widely referred to as "Liberation Day", the United States Government announced broad-based tariffs on virtually all of its trading partners, citing persistent trade deficits and national security concerns. The policy included a base 10% tariff and, subsequently, additional higher rates for trading partners with large surpluses against the U.S., such as China and the European Union. Following the announcement, the dollar fell sharply against a broad basket of currencies, with the Euro, Yen and Swiss Franc appreciating against it. The move was accompanied by a spike in equity-market volatility and a U.S. equities sell-off.

Less than a year later, on 28 February 2026, the United States and Israel launched airstrikes on Iran, beginning the 2026 Iran war. As escalation in the Gulf threatened to close the Strait of Hormuz and sent crude sharply higher, the dollar did exactly what the safe-haven literature predicts and appreciated.

This thesis speaks to a live debate about whether the dollar's exceptional status is eroding. A growing literature argues that the currency's safe-haven premium has weakened, pointing to the unusual co-movement of a falling dollar, rising Treasury yields and rising risk premia during the 2025 tariff episode ([Kamin 2025](#); [Jiang et al. 2026](#)). Against this, the long tradition on the dollar's "exorbitant privilege" and its role as the world's reserve and safe-haven currency ([Gourinchas and Rey 2005](#); [Obstfeld and Zhou 2023](#); [Habib and Stracca 2012](#); [De Bock and de Carvalho Filho 2015](#)) would lead one to expect the status to persist. The two 2025–2026 shocks offer an unusually clean natural experiment for adjudicating the debate, because they differ in *kind*: one is a U.S. trade-policy shock, the other a geopolitical oil-supply shock.

The central question of this thesis is therefore: *does the dollar's safe-haven response depend on the type of shock, even when the United States is involved in both?* To adjudicate it, the design sets the two shocks against a historical benchmark, giving three episodes in all. The Liberation Day tariff announcement is a purely U.S.-originated economic-policy shock. The 2026 Israel–Iran/Hormuz crisis is a geopolitical oil-supply shock in which the United States is involved as a co-belligerent but still functions as a financial safe harbour. And the 2022 Russian invasion of Ukraine, an external military/oil shock not authored by the United States, serves as a control for the "classical" safe-haven response. Comparing the three isolates whether the dollar's behaviour is conditioned by the nature of the shock or merely

by who caused it.

The thesis answers the question with two complementary empirical layers on a single daily dataset spanning January 2019 to June 2026. An *event study* measures the cumulative abnormal response of the dollar, a cross-section of currency baskets, gold, equities and crude oil around each shock, and tests formally whether the dollar's response differs across events. A complementary *regression* then tests whether the dollar's relationship with risk itself changed across the episodes by examining whether its safe-haven sensitivity to volatility held or broke. The two layers are complementary: the event study isolates the dollar's move around each discrete, dated shock, while the regression tests whether its underlying sensitivity to risk shifted across the episodes.

The contribution is threefold. First, to the best of the author's knowledge this is the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets. Second, the thesis bridges the safe-haven and commodity-currency literatures by carrying the oil channel and a gold benchmark through the same design, so that the dollar's behaviour can be read against gold and the oil-exposed currency cross-section rather than in isolation. Third, a 32-currency spot panel from LSEG (London Stock Exchange Group), supplemented by one-month forward quotes for 28 currencies and built for this thesis, carries the standard currency baskets through to mid-2026, so the most recent shocks can be studied with consistent, citable data.

The remainder of the thesis is organised as follows. Chapter 2 reviews the safe-haven, dollar-erosion and commodity-currency literatures. Chapter 3 sets out the conceptual framework, including the working definition of a safe-haven currency, a classification that separates a shock's origin from its nature, and the hypotheses that follow. Chapter 4 narrates the 2022–2026 events. Chapter 5 describes the data and the event-study and regression methodologies. Chapter 6 presents the empirical results. Chapter 7 discusses the findings and their implications, and Chapter 8 concludes.

The results point to a dollar whose safe-haven status is conditional on the type of shock rather than on its originator. The two U.S.-involved shocks produce a mirror image: the Liberation Day tariffs depreciate the broad dollar index by over three and a half percent within twenty trading days, while the Hormuz oil-supply crisis appreciates it by an amount comparable to the external Ukraine benchmark at the five-day horizon (+1.53 against +1.03 percent), from which its response is statistically indistinguishable at five and ten days. The complementary regression

1 Introduction

supports the pattern. The dollar's risk sensitivity, significantly positive on normal days, collapses to essentially zero inside the tariff window yet holds through the Hormuz and Ukraine episodes. Gold emerges as a hedge against the dollar rather than a universal safe haven, and the clearest exporter–importer separation occurs after the tariff shock, when crude fell, while the Hormuz spread remains insignificant and Ukraine provides weaker evidence at twenty days.

2 Literature Review

This chapter situates the thesis within six distinct strands of literature. Each body of research addresses a specific dimension of the core question the thesis tries to answer: does the dollar’s status as a safe haven depend on the nature of a shock, rather than its source? By reviewing these fields, this chapter establishes the theoretical framework to understand how the dollar functions as a safe haven, setting the stage for examining the consequences of when the United States itself originates a shock, acts as a co-belligerent or is not directly involved.

2.1 Currency Risk Factors

[Lustig et al. \(2011\)](#) establish the foundational framework for understanding currency risk by identifying two primary common risk factors, a “level” factor, the average excess return on foreign-currency portfolios against the dollar, and a “slope” factor, which captures the excess returns of carry trades. They demonstrate that high-interest-rate currencies load more heavily on the slope factor, which is closely linked to global equity-market volatility. Crucially, their model shows that during periods of high global risk, low-interest-rate currencies endogenously appreciate, acting as a hedge, while high-interest-rate currencies depreciate.

The dollar’s special role is not confined to asset markets. It also dominates the invoicing of international trade. Drawing on evidence from more than 2,500 country pairs covering some ninety per cent of world trade, [Gopinath et al. \(2020\)](#) show that traded-goods prices are predominantly sticky in dollars rather than in the exporter’s or the importer’s currency. Trade prices and volumes therefore respond to the dollar exchange rate more than to the bilateral rate.

Building on this dominant-currency paradigm, [Dalgic and Ozhan \(2025\)](#) show that a currency’s risk premium is not exogenous but rooted in an economy’s trade and financial structure, namely the share of its trade invoiced in dollars and the amount of dollar debt it carries. In their small open-economy model, a shock that depreciates the local currency, such as a foreign interest-rate hike, turns contractionary rather than expansionary. Dollar debt opens a financial channel through which the depreciation raises the real value of liabilities. Servicing dollar debt now requires more local currency. Dollar invoicing mutes the trade channel, leaving export volumes largely unresponsive to the exchange rate, providing no offset. Because depreciations are therefore recessionary, the exchange rate turns countercyclical. The currency loses

value precisely when domestic income falls, offers its holder no insurance, and is therefore a poor hedge against domestic downturns. Investors hold such currencies only against a sufficient risk premium, the mirror image of the safety commanded by the dollar. The dollar and carry factors, extended to June 2026, form the basis of the portfolio construction in Chapter 5.

2.2 Dollar Dominance and Its Erosion

The dollar’s role as the primary global reserve currency confers an “exorbitant privilege”, a persistent excess return on U.S. external assets over its liabilities (Gourinchas and Rey 2005). [Gourinchas et al. \(2017\)](#) show that this privilege is mirrored by an “exorbitant duty”. In global crises the United States transfers wealth abroad, roughly 19% of U.S. gross domestic product (GDP) during 2007–2009. Because it holds risky foreign assets while owing safe dollar liabilities, collapsing global asset prices and the appreciating dollar both push its net foreign asset position downward, with the exchange rate accounting for roughly a third of the valuation effect. [Jiang et al. \(2021\)](#) find the source of that movement, showing that surges in foreign demand for safe dollar assets appreciate the dollar immediately. They capture the demand through the convenience yield on U.S. Treasuries, the return investors give up in exchange for safety and liquidity. The dollar’s appreciation in a global crisis is therefore a consequence of its role as the world’s supplier of safe assets. [Obstfeld and Zhou \(2023\)](#) add the cyclical dimension, describing a “global dollar cycle” in which dollar appreciation tightens financial conditions worldwide. They find that dollar movements correlate most reliably with measures of investor risk appetite.

While the 2008 financial crisis and the COVID-19 pandemic elicited the standard appreciation, [Jiang et al. \(2026\)](#) argue that the dollar has been losing that standing gradually, documenting convenience yields on dollar safe assets eroding over the two years before the 2025 tariff announcements. A comparable trend is evident in central bank reserves. [Arslanalp et al. \(2022\)](#) document a two-decade decline in the dollar’s share of global reserves, and attribute it to deliberate diversification by reserve managers rather than to valuation effects. Between 1999 and 2021 the dollar’s share of global reserves declined from 71% to 59%. A quarter of what left the dollar moved into the Chinese renminbi, the remaining three quarters into currencies of smaller economies that had rarely served as reserve assets before.

2.3 Safe-Haven Currencies and Gold

A central paradox, identified by [Habib and Stracca \(2012\)](#), is the dollar's appreciation during the 2008 global financial crisis, a crisis that originated in the United States. Analysing 52 currencies over almost 25 years, they find that safe-haven status is best predicted not by the interest-rate spread that the carry-trade literature emphasises but by a country's net foreign asset position, which signals how exposed it is to external stress. A currency that has hedged well in the past also tends to continue doing so, and the currencies of large but less financially open economies hedge better in periods of distress. The interest-rate spread against the United States, by contrast, matters only during the 2008 crisis itself.

[Cheema et al. \(2022\)](#) compare a range of safe-haven assets across different crises, and find their safe-haven character changed considerably during the COVID-19 pandemic compared with the 2008 financial crisis. The Swiss franc remained an intermediate haven in most settings, while gold and silver, weak havens in 2008, turned risky. The dollar deteriorated as well, from intermediate protection in 2008 to weak protection during the pandemic.

Providing broader context, [De Bock and de Carvalho Filho \(2015\)](#) demonstrate that the dollar, Japanese yen and Swiss franc consistently appreciate against other currencies during risk-off episodes. How much each currency gains depends on its interest rate before the episode, its net foreign asset position, whether it is overvalued, and how cheaply and freely it can be traded. [Ranaldo and Söderlind \(2010\)](#) show that during U.S. equity sell-offs the franc and yen appreciate against the dollar, with effects that appear within hours, persist for several days, and grow more than proportionally as volatility rises. The dollar therefore strengthens broadly in risk-off episodes while still losing ground to the franc and the yen.

[Baur and Lucey \(2010\)](#) are the first to formally define and test the distinction between a hedge and a safe haven. In their terminology, a hedge is uncorrelated or negatively correlated with equities on average, which does not guarantee protection during market crashes. A safe haven must deliver exactly that protection in the extreme. Applying the distinction to gold, they find that it hedges equities on average, but works as a safe haven only after severe market downturns, and even then only for a short time period. This strand supplies the thesis's working definition of a safe haven and its benchmarks. Chapter 3 adopts the conditional, crisis-contingent notion of safety, and the yen, the franc and gold serve as the standards against which the dollar's response is measured.

2.4 Geopolitical Risk

Caldara and Iacoviello (2022) provide the foundational measurement of geopolitical risk, a news-based index that separates geopolitical “threats”, events that are feared but have not yet occurred, from geopolitical “acts”, their actual realisation. Shocks to the index depress investment, employment and stock prices, and the damage persists, with both the threat and the realisation components contributing. Expanding this to exchange rates, Yilmazkuday (2025) demonstrates that the dollar’s geopolitical safe-haven status is not universal. A year after a shock to global geopolitical risk, the U.S. dollar has depreciated significantly, even as currencies such as the South African rand or Brazilian real have appreciated. The currencies of countries most deeply integrated into international supply chains depreciate most, a pattern driven primarily by geopolitical acts rather than threats. Complementing this, Liu and Zhang (2024) find that a zero-cost strategy that buys the currencies of high-geopolitical-risk countries and sells those of low-risk countries earns a significant excess return of 5.72% per year, and that this return predictability stems from country-specific and regional risk components rather than a single global factor. A currency’s risk–return profile is therefore sensitive to the geographic origin of the shock.

Institutional evidence points in the same direction. The International Monetary Fund (2025) finds that although geopolitical risk events exert only a modest effect on asset prices on average, major events, military conflicts in particular, produce substantially larger and more persistent equity declines and raise governments’ borrowing costs. The effects are strongest in emerging markets with little fiscal room and limited foreign reserves, and they spill over to other countries through trade and financial ties.

The thesis draws on this strand directly. The aggregate index places the Hormuz and Ukraine episodes in their longer-run context in Chapter 4, and changes in the index serve as a measure of geopolitical risk in the safe-haven regression.

2.5 Tariff Shocks and Exchange Rates

The 2025 tariff episode has generated a fast-growing literature that is central to this thesis. Kamin (2025) estimates the dollar’s daily relationship with the VIX, the Chicago Board Options Exchange’s index of expected S&P 500 volatility, and with interest differentials and finds a significant break after the Liberation Day announcements of April 2025. Its sensitivity to the VIX turns from positive to

negative, so a currency that once gained when volatility spiked now falls with it, and investors no longer treated the dollar as a refuge in stress. [Jiang et al. \(2026\)](#) document the same break across a wider set of correlations. The dollar depreciated while U.S. interest rates rose relative to euro rates, the VIX climbed and the convenience yield on Treasuries fell, a pattern they attribute to weakening foreign demand for the safe assets the United States supplies. [Hassan et al. \(2025\)](#) provide the theoretical counterpart in a general-equilibrium model where a currency's safety arises endogenously. In their account the dollar is safe because disturbances originating in the United States move traded-goods prices worldwide, and a trade war that cuts the country off from world markets shuts down those spillovers, so the dollar's habit of strengthening in bad times fades and U.S. interest rates move higher.

The depreciation was not what the earlier record suggested. [Khalil and Strobel \(2024\)](#) show that the tariff rounds of 2018–2019 came with a stronger dollar and trace that appreciation to U.S. trade policy itself. [Ostry et al. \(2025\)](#) qualify the pattern with a database of tariff announcements, threats and implementations spanning 2018–2020 and 2025, weighting each shock by the size of the tariff and its economic relevance. The direction of the response, they find, hinges on retaliation. The dollar appreciates when the United States imposes tariffs unilaterally but depreciates when trading partners strike back, which leads them to conclude that the fall after 2 April 2025 was no anomaly once retaliation is taken into account. [Kalemli-Özcan et al. \(2026\)](#) offer a complementary explanation centred on uncertainty. In their model, doubt about future tariff policy feeds a risk premium into the exchange rate, and greater tariff uncertainty raises precautionary saving and the compensation investors demand, weakening the currency at once even though the tariffs alone would strengthen it. Calibrated to the 2025 episode, the model matches how far and how fast the dollar fell.

[Benguria and Saffie \(2025\)](#) measure the immediate market response to the 2 April announcement itself, separating the anticipated part of each country's tariff from the surprise created by rounding the rates upward. A one-percentage-point higher surprise tariff lowers the local-currency equity prices of listed firms in the affected trading-partner countries by 0.23%, and the anticipated part also lowers them, significantly but by an order of magnitude less. The dollar shows no appreciation, if anything a depreciation against floating currencies. This strand supplies the thesis's first treatment. The Liberation Day shock enters the event study as the trade-policy episode against which the military and oil-supply shocks are compared.

2.6 Commodity Currencies and Oil

Y.-C. Chen and Rogoff (2003) show that for commodity exporters such as Australia, Canada and New Zealand, the world price of their exports is a strong driver of the real exchange rate. These three serve as the natural test cases, developed economies with floating exchange rates whose exports remain dominated by commodities. The effect is clearest for the Australian and New Zealand dollars. Kilian (2009) decomposes oil-price changes into supply disruptions, global demand for industrial commodities, and precautionary demand reflecting concern about future supply, each with distinct effects on the oil price, output and inflation. A supply-driven price increase is therefore not equivalent to a demand-driven one.

H. Chen et al. (2016) show that the dollar reacts more strongly to demand-driven than to supply-driven oil shocks. The reaction is largest against the Canadian dollar and the Norwegian krone, because Canada and Norway trade oil heavily with the United States. Oil has also become more important for the dollar over time. Before the global financial crisis, oil shocks accounted for 10% to 20% of the movement in the dollar's bilateral rates. Afterwards the share rises to over 40%, and for the Canadian dollar to more than half.

Albulescu and Ajmi (2021) document predictive causality between oil prices and the dollar in both directions, strongest in crisis periods. Salisu et al. (2022) show that oil tail risk predicts the tail risk of the dollar's rates against the Canadian dollar and the British pound, while the dollar-yen pair moves inversely, consistent with the yen's haven role.

For the thesis, this literature provides the basis for the oil-exporter versus oil-importer portfolio split in Chapter 5, and the distinction between supply-driven and demand-driven oil movements (Kilian 2009) separates the Hormuz crisis from the tariff shock. The episodes themselves, and the sources that document them, are covered in Chapter 4.

3 Conceptual Framework

3.1 The Dollar as a Safe Haven

In periods of high global risk investors reallocate from risky assets toward safe assets, meaning assets they expect to hold value. Global risk is high when equity markets fall and expected volatility rises. For the dollar this demand concentrates in U.S. Treasuries, whose safety and liquidity command a premium. The resulting surge in demand for dollar safe assets appreciates the dollar ([Jiang et al. 2021](#)).

The safe-haven channel connects two observable variables. Global risk raises the demand for safe assets and is measured with the VIX. The dollar's return shows whether that demand appreciates the dollar and is measured on the broad dollar index and the currency baskets of Chapter 5. Following [Baur and Lucey \(2010\)](#), an asset is a hedge if it is uncorrelated or negatively correlated with risky assets on average, and a safe haven if that protection holds in times of market stress. The dollar therefore behaves as a safe haven if its return rises when equity prices fall and volatility increases ([Ranaldo and Söderlind 2010](#); [De Bock and de Carvalho Filho 2015](#)).

The channel rests on three assumptions. First, exchange rates incorporate public news within days, so a short event window can attribute a price move to the event. Second, the VIX is a valid measure of global risk appetite. Third, around large shocks the dollar's short-window response reflects the demand for safety rather than changing interest-rate expectations. Under these assumptions the channel makes a sign prediction. In the days after the announcement of a large adverse shock, the dollar's cumulative abnormal return from day 0 to day h , $CAR(0, h)$ as defined in Section 5.2, should be positive,

$$CAR(0, h) > 0 \quad \text{for short horizons } h. \quad (3.1)$$

The event study of Chapter 5 tests exactly this sign. The same logic implies that on ordinary days the dollar's return should rise with the VIX, a positive relation the regression checks as supporting evidence. The origin of a shock appears nowhere in the channel. On its own it therefore predicts a positive abnormal return around every risk-off event, even one that originates from U.S. policy itself.

3.2 The Limits of Dollar Safety

The dollar's value includes a premium for safety. Investors accept a lower return on U.S. safe assets, the convenience yield, and this premium is priced into the dollar's exchange rate (Jiang et al. 2021). The premium rests on the expectation that dollar assets will remain safe. A shock that originates in U.S. policy can lower exactly this expectation. The demand for dollar safe assets then falls despite rising global risk, and the dollar depreciates (Hassan et al. 2025). Around a U.S.-originated trade-policy shock this erosion channel therefore predicts the opposite sign on the same quantity, a negative cumulative abnormal return in the days after the announcement.

The erosion channel runs through the same two variables as the safe-haven channel, global risk and the dollar's return, and adds the safety premium as a third. It rests on two assumptions of its own. First, the dollar's value reflects expected future convenience yields and not only current trade flows and interest rates, so news alone can move the dollar (Jiang et al. 2021). Second, markets read a U.S. trade-policy shock as news about the future quality of dollar assets and not only as news about trade (Hassan et al. 2025). Under these assumptions the sign prediction of the safe-haven channel flips for exactly one type of event. Around the announcement of a shock that originates in U.S. policy, the erosion channel predicts

$$\text{CAR}(0, h) < 0 \quad \text{for short horizons } h \quad (3.2)$$

while the safe-haven channel alone would predict the opposite. Inside such an episode the dollar's usual positive relation to the VIX should also weaken or vanish, which the regression examines.

Gold separates the two channels. Its role as a crisis haven for equities exposes it to the safe-haven channel (Baur and Lucey 2010). At the same time it has no issuer, so no government's policy stands behind its value and the erosion channel cannot operate on it. Around a U.S.-originated trade-policy shock the two assets should therefore diverge. The dollar carries both channels and can fall, while gold carries only the safe-haven channel and should hold or rise. If the dollar loses its safe-haven response only when the shock is U.S.-originated while gold does not, the failure lies in the erosion channel and not in a general flight from safe assets.

3.3 Design of the Comparison

Because both channels respond to risk, a single episode cannot tell them apart. Separating them needs variation in two attributes of a shock, who originates it and what kind of shock it is. The two attributes have four combinations, and the three episodes studied fill three of them. The 2025 Liberation Day tariffs are a U.S.-originated trade-policy shock. The 2026 Israel–Iran/Hormuz crisis is a military/oil-supply shock in which the United States is involved as a co-belligerent. The 2022 invasion of Ukraine is an external military/oil-supply shock that the United States did not author. The fourth combination, an external trade-policy shock, has no counterpart in the sample. Chapter 4 sets out the three episodes and the sources that document them. If only the safe-haven channel operates, all three episodes should appreciate the dollar and origin should not matter. If the erosion channel also operates, the U.S.-originated trade-policy shock should stand apart while the two military/oil-supply shocks resemble each other regardless of U.S. involvement. The two channels thus make opposite predictions about which episodes cluster together.

The design has limits. Each occupied combination holds a single episode. The two U.S.-linked shocks differ in the degree of involvement, as originator in one case and co-belligerent in the other. The fourth combination stays empty, so no purely external policy benchmark exists. The comparison therefore identifies patterns across these three episodes rather than properties of shock classes.

The oil market supplies both a classifier and a third channel. In the decomposition of [Kilian \(2009\)](#), the Hormuz episode is broadly supply-side, lifting crude through disrupted shipments and through precautionary demand about future supply, whereas the tariff shock depresses crude through weaker expected demand. The sign of the crude move therefore marks the nature of a shock. Through the commodity-currency channel the same move should also sort currencies, oil exporters outperforming importers when crude rises and underperforming when it falls ([Y.-C. Chen and Rogoff 2003](#); [H. Chen et al. 2016](#)).

The cross-section is read on the currency baskets of Chapter 5, the funding and carry portfolios ([Lustig et al. 2011](#)), the literature-standard baskets and the split into oil exporters and importers. Gold is read in dollars and in euros, so that mechanical denomination effects do not drive its measured response.

3.4 Hypotheses

The two channels lead to one overall prediction. The dollar's safe-haven response depends on the nature of a shock, not on U.S. involvement. Three ex-ante hypotheses make this testable.

The first hypothesis, H1, holds that responses cluster by the nature of a shock, not by U.S. involvement. Around the military/oil-supply shocks, Ukraine and Hormuz, the safe-haven channel governs and the dollar appreciates in both, whatever the degree of U.S. involvement. Around the U.S.-originated trade-policy shock the erosion channel governs and the dollar depreciates. The content of H1 is the joint pattern across the three episodes, not any single response.

The second hypothesis, H2, makes gold the origin-blind benchmark. Gold is exposed to the safe-haven channel but not to the erosion channel, so its response does not discriminate between shocks by origin. It holds or rises even in the U.S.-originated episode, in which the erosion channel predicts a weaker dollar. Any conditionality under H1 is then specific to the dollar.

The third hypothesis, H3, carries the oil channel into the cross-section. The exporter–importer spread follows the sign of the crude move. Oil exporters outperform importers when a shock lifts crude, as under Hormuz and Ukraine, and lose ground when it depresses crude, as on Liberation Day.

These predictions are not foregone, since [Yilmazkuday \(2025\)](#) finds the dollar depreciating in the year after global geopolitical-risk shocks. H1 fails if the tariff episode moves like the military pair or if the military pair splits along U.S. involvement. H2 fails if gold weakens in the U.S.-originated episode. H3 fails if the spread ignores the sign of the crude move. Chapter 6 reports the tests and Chapter 7 returns the verdicts.

4 Background

This chapter documents the episodes the empirical chapters analyse and establishes the dates that define the event windows of Chapter 5. For each episode it sets out what happened and how official institutions described the market reaction.

4.1 Liberation Day

On 2 April 2025, in an announcement branded “Liberation Day,” the United States imposed broad “reciprocal” tariffs on nearly all of its trading partners, combining a flat baseline rate of ten percent with higher country-specific surcharges. The measure was imposed under the International Emergency Economic Powers Act, after the President declared that large and persistent trade deficits and a lack of reciprocity in trade relationships amounted to a national emergency. It applied to most U.S. imports of goods ([Congressional Research Service 2026a](#)). Later assessments found that foreign exporters absorbed only about five percent of the tariff cost, so almost all of it fell on U.S. firms and consumers, while import volumes declined sharply ([European Central Bank 2026](#)).

Equity prices fell sharply and implied volatility spiked to a level exceeded only during the global financial crisis and the COVID-19 pandemic ([European Central Bank 2025](#)), while crude oil fell as markets priced slower growth and weaker oil demand ([U.S. Energy Information Administration 2026a](#)). Around the announcements the dollar depreciated against the other major advanced-economy currencies even as risk metrics deteriorated and Treasury yields rose ([Kamin 2025](#)). A partial ninety-day pause followed on 9 April 2025, suspending the country-specific surcharges while the baseline rate stayed in force ([Congressional Research Service 2026a](#)).

The legal basis of the tariffs remained contested. Multiple federal courts ruled that the President had exceeded his authority in at least some of the actions. The tariffs stayed in force while the Supreme Court reviewed the cases ([Congressional Research Service 2026a](#)). On 20 February 2026 the Supreme Court held that the act does not authorise the President to impose tariffs ([Supreme Court of the United States 2026](#)). Tariffs imposed under other trade statutes were not before the Court.

4.2 Twelve-Day War and Hormuz Crisis

Israel struck Iran on 13 June 2025, opening a twelve-day war that drew in U.S. strikes on three Iranian nuclear sites on 22 June. It ended on 24 June, hours after an Iranian missile strike on a U.S. base in Qatar ([Congressional Research Service 2025](#)). The episode lifted crude only briefly ([Kilian et al. 2026](#); [U.S. Energy Information Administration 2026a](#)).

The conflict resumed on 28 February 2026, when the United States and Israel opened a joint campaign of airstrikes on Iran ([Congressional Research Service 2026b](#)). Both governments said they aimed to end Iran’s nuclear and missile capability and to force a change of government in Tehran. Iran’s response was regional, reaching Israel, U.S. military sites across the Middle East, and the Arab states that host American forces, among them Bahrain, Kuwait and Qatar. Iran also announced that it would close the Strait of Hormuz ([House of Commons Library 2026a](#)), which carries roughly a quarter of globally traded oil ([Fattouh and Economou 2026](#)). Attacks on shipping and the repricing of tanker insurance effectively closed the Strait within days ([Kilian et al. 2026](#)). On 13 March 2026 the United States reportedly struck military targets on Kharg Island, the terminal through which some ninety percent of Iran’s oil exports transit ([Congressional Research Service 2026b](#)). Brent crude, which had traded near \$71 per barrel before the 28 February strikes, rose by 46 percent over March to a monthly average of about \$104 and, in front-month futures, peaked at \$118 on 29 April ([Fattouh and Economou 2026](#); [International Monetary Fund 2026](#); [U.S. Energy Information Administration 2026c](#)). The daily spot series plotted in Figure 4.2 reached \$138 on 7 April ([U.S. Energy Information Administration 2026b](#)).

A two-week ceasefire announced on 7 April 2026 came with an Iranian declaration that the Strait was open, though only to vessels using a route coordinated with Iranian port authorities, which falls short of free passage ([Fattouh and Economou 2026](#)). Talks in Islamabad on 11 and 12 April broke down, and the United States began a naval blockade of Iranian ports on 13 April. The ceasefire itself held into May, but Iran resumed attacks on 4 May and the United States struck Iran again on 25 May ([Congressional Research Service 2026c](#); [CNN 2026](#)). A memorandum of understanding signed on 17 June set out plans for talks over an initial sixty-day period on freedom of navigation, Iran’s nuclear and missile programmes and sanctions ([House of Commons Library 2026b](#)). Once it was signed and tanker movements picked up, Brent fell through the rest of the quarter to \$72 on 26 June ([U.S. Energy Information Administration 2026c](#)).

The fighting did not stop with the agreement. Between 25 and 28 June the United States struck Iran in response to an attack on commercial shipping in the Strait, and Iran struck U.S. bases in Bahrain and Kuwait ([House of Commons Library 2026b](#)). The sample analysed in this thesis ends on 30 June 2026, with the conflict unresolved.

4.3 Russian Invasion of Ukraine

On 24 February 2022 Russian forces crossed the border into Ukraine and began a full-scale assault on the country. The European Union, the United Kingdom, the United States and other allies answered within days with coordinated sanctions targeting Russia's banks, its aviation and shipping sectors, and industries of strategic significance to the Russian state, among them defence, aerospace and energy ([House of Commons Library 2025](#)).

Two measures reached the international payments and reserve system directly. On 26 February 2022 the European Union, the United Kingdom, the United States and Canada announced that they would restrict the Russian Central Bank's access to its foreign reserves. The stated aim was to undercut its ability to conduct foreign exchange transactions in support of the rouble. Russia was thought to hold more than \$630 billion in currency and gold reserves at the time. By late 2023 more than sixty percent of them had been frozen. On 2 March 2022 the European Union announced the exclusion of seven Russian banks from the SWIFT (Society for Worldwide Interbank Financial Telecommunication) payments network, effective ten days later. Three more followed in June ([House of Commons Library 2025](#)).

Energy markets had already been pricing the risk of an invasion. Crude oil rose and grew more volatile from mid-January 2022 onward, passed \$100 per barrel as the attack began, and Brent futures closed just below \$115 per barrel on 2 March ([U.S. Energy Information Administration 2022](#)).

4.4 Overview of the Events

The analysis dates six events. The two treatments are Liberation Day, a trade-policy shock the United States authored, on 2 April 2025, and the Hormuz crisis, a military and oil-supply shock in which it was a belligerent, on 28 February 2026. Each is followed by a reversal within weeks, the ninety-day tariff pause on 9 April 2025 and the two-week ceasefire on 7 April 2026. The invasion of Ukraine on 24 February 2022 enters as the external control, a shock the United States did not author, and the

4 Background

twelve-day war of 13 June 2025 as a second, milder military shock.

Figures 4.1–4.3 place these events in their longer-run context, plotting the broad dollar index, the price of Brent crude, and the geopolitical-risk index of [Caldara and Iacoviello \(2022\)](#) over 2019–2026, with the four main events marked on each series. Chapter 6 measures the responses formally.

4 Background



Figure 4.1: Broad dollar index, 2019–2026. Source: [Board of Governors of the Federal Reserve System \(2026c\)](#), via FRED (Federal Reserve Economic Data).



Figure 4.2: Brent crude oil price, 2019–2026, USD per barrel. Source: [U.S. Energy Information Administration \(2026b\)](#), via FRED.

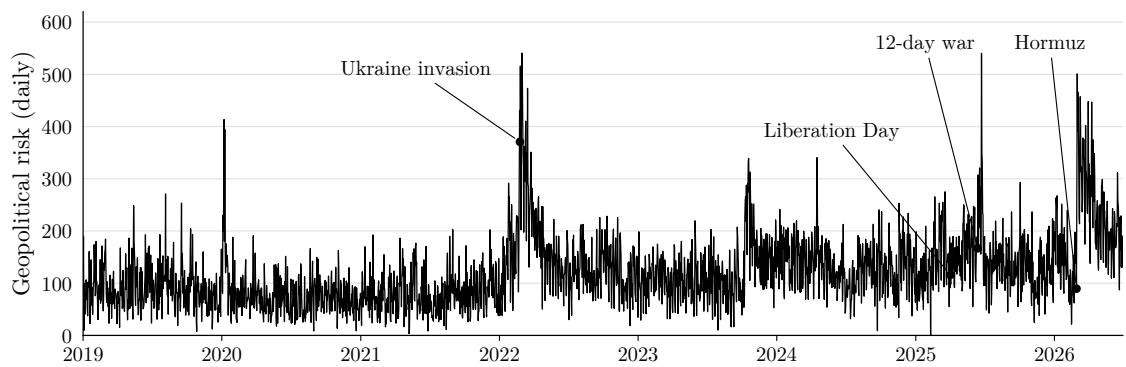


Figure 4.3: Daily Geopolitical Risk Index, 2019–2026. Source: [Caldara and Iacoviello \(2026\)](#).

5 Data and Methodology

This chapter describes the data and the empirical methods used for the analysis in Chapter 6. Section 5.1 describes the exchange rates and financial variables, the geopolitical and policy uncertainty indices, and the construction of the currency baskets. Section 5.2 sets out the event study and Section 5.3 the regression.

5.1 Data

5.1.1 Exchange Rates and Financial Variables

The currency panel contains daily exchange rates for 32 currencies against the U.S. dollar from 1 January 2019 to 30 June 2026. Spot rates are the mid of the closing bid and ask quotes from LSEG Workspace ([LSEG 2026d](#)), with the Federal Reserve’s H.10 release as a cross-check ([Board of Governors of the Federal Reserve System 2026a](#)). All rates are expressed as foreign currency per dollar, and the four that LSEG quotes as dollars per currency (EUR, GBP, AUD, NZD) are inverted. Daily returns are computed as log changes,

$$r_{i,t} = \ln(S_{i,t-1}/S_{i,t}), \quad (5.1)$$

where $S_{i,t}$ is the spot rate of currency i on day t . A positive return is an appreciation of the foreign currency against the dollar.

The remaining series come from FRED and LSEG. FRED provides the nominal broad dollar index, the 10-year Treasury yield, the VIX, the Brent and West Texas Intermediate (WTI) crude prices and the S&P 500 index ([Board of Governors of the Federal Reserve System 2026c](#); [Board of Governors of the Federal Reserve System 2026b](#); [Cboe Global Markets 2026](#); [U.S. Energy Information Administration 2026b](#); [S&P Dow Jones Indices 2026](#)). Brent serves as the global crude benchmark and WTI as the U.S. benchmark ([U.S. Energy Information Administration 2022](#)). LSEG provides the gold price in dollars and the Euro Stoxx 50 index in euros ([LSEG 2026a](#)). Series identifiers are listed in Chapter 10. Indices, equities and commodities enter as log returns. The 10-year yield and the VIX enter in first differences.¹

¹Missing prices over market holidays are filled with the previous quote for up to three days, producing exact zero returns on 3.5 percent of Brent trading days, 0.6 percent of gold days and 4.3 percent of days of the broad dollar index. A few such days fall inside event windows (Good Friday and Easter Monday 2025 in the Liberation Day window, Juneteenth and Independence Day 2025 in the twelve-day-war window), and no estimation window holds more than nine zero

Because oil and gold are priced in dollars, part of any change in their dollar price reflects the dollar itself rather than the commodity. A depreciation of the dollar raises the dollar price of oil even when oil is unchanged in every other currency. To separate commodity movements from this denomination effect, Brent and gold are additionally re-expressed in euros as

$$r_{X,t}^{\text{EUR}} = r_{X,t}^{\text{USD}} - r_{\text{EUR},t}, \quad (5.2)$$

where X denotes Brent or gold and $r_{\text{EUR},t}$ is the euro's log appreciation against the dollar. The euro-denominated series enter the event study as a robustness check on denomination.

5.1.2 Geopolitical and Policy Uncertainty Indices

Geopolitical risk is measured by the daily Geopolitical Risk (GPR) index of [Caldara and Iacoviello \(2022\)](#) and trade-policy uncertainty by the daily Trade Policy Uncertainty (TPU) index of [Caldara et al. \(2020\)](#), both downloaded from the authors' site ([Caldara and Iacoviello 2026](#); [Caldara et al. 2026](#)).² Both indices count newspaper coverage. The GPR index measures the share of articles discussing adverse geopolitical events in ten leading U.S., British and Canadian newspapers, and the TPU index the share of articles discussing trade-policy uncertainty in seven major newspapers. The J.P. Morgan VXY Global index of implied U.S.-dollar exchange-rate volatility is the currency-market counterpart of the VIX ([LSEG 2026b](#)). The one-month forward discount $f_{i,t} - s_{i,t}$, the log forward minus the log spot rate, with the forward rates from LSEG Workspace ([LSEG 2026c](#)), proxies the interest-rate differential under covered interest parity ([Lustig et al. 2011](#)) and is the sorting variable for the currency baskets of Section 5.1.3.

5.1.3 Portfolio Construction

The safe-haven prediction concerns types of currencies rather than any single one, so the currencies enter the event study in baskets. The central pair, a safe and a risky basket, is defined in two ways. In the first, currencies are sorted into terciles

days for any of the price series reported here. The 10-year yield has more zero days because it is quoted to two decimals and can be genuinely unchanged. Recomputing the headline cross-event statistic without the zero days moves its t from -4.11 to -4.0 , so the effect on the reported statistics is negligible.

²The TPU series is revised as it is updated, so the results reflect the version downloaded on 23 July 2026.

by their average one-month forward discount over the sample. The low-yield tercile forms the safe basket and the high-yield tercile the risky basket, with pegged and managed currencies (DKK, HKD, SAR, KWD, CNY) excluded from the sort.³ Four further currencies (BRL, CLP, COP, PEN) trade forward only as non-deliverable contracts without a clean one-month quote and drop from the currency sort for lack of data, leaving 23 currencies in the sort. The choice of terciles follows the sizing convention of the sorting literature, which builds fewer, broader portfolios when there are few assets to sort. [Asness et al. \(2013\)](#) sort the ten currencies in their sample into three equal groups and [Lustig et al. \(2011\)](#) form six portfolios from at most 35 currencies. Sorting the 23 currencies into terciles gives the safe and the risky basket seven each. Quintiles would leave four or five, few enough that a single currency's swings could dominate the basket. This fixed tercile classification defines the safe and risky currency baskets used in the main event-study specification (the full classification is given in Table 1 in the Appendix). Because the low-yield tercile also contains low-rate Asian currencies that are not conventional safe havens, a second, literature-standard scheme is reported as a robustness check, pairing the yen, Swiss franc and euro as the safe basket ([Ranaldo and Söderlind 2010](#)) with the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah as the risky basket ([De Bock and de Carvalho Filho 2015](#)). The euro enters the literature-standard safe basket as the largest liquid alternative to the dollar rather than as a documented safe haven. Direct haven evidence for the euro is confined to the period before 2008 ([Habib and Stracca 2012](#)), but recent work on the dollar's reserve status treats the euro as the principal alternative currency ([Jiang et al. 2026](#)). A separate split tracks the oil channel. Currencies are grouped by their net crude oil trade over 2019–2024 into oil exporters (NOK, CAD, MXN, COP, BRL, SAR, KWD) and oil importers (JPY, KRW, INR, THB, TWD, TRY) ([United Nations Statistics Division 2026](#))⁴. [H. Chen et al. \(2016\)](#) apply the same split to the six of these currencies in their sample, grouping Canada, Norway and Mexico as oil exporters and Japan, Korea and Turkey among the importers. A higher oil price improves the exporters' terms of trade and worsens the importers', so the two baskets should separate in oil-driven shocks ([Y.-C. Chen and Rogoff 2003](#); [H. Chen et al. 2016](#)). All baskets are equally weighted, the convention of the sorting literature.

³HKD, SAR and KWD are fixed to the dollar or to a dollar-dominated basket and CNY is tightly managed, so against the dollar these four cannot move freely on risk news. DKK is pegged to the euro instead: against the dollar it moves almost one-for-one with the euro (correlation 0.998 over the sample) and is excluded as a euro duplicate rather than for lack of movement.

⁴Türkiye reports no crude-oil trade to UN Comtrade, so its importer status is measured from the mirror side, the crude exports of all other countries to Türkiye.

The dollar basket averages the dollar’s return against all 32 currencies.

5.2 Event Study Methodology

The main specification is the constant-mean return model ([MacKinlay 1997](#); [Campbell et al. 1997](#)). The event window is $[-20, +20]$ trading days. For each series i the normal return is its average over an estimation window of $[-140, -21]$ trading days preceding the event, ending where the event window begins so that no event-window day enters the estimation of the benchmark. The abnormal return on day t of the event window is the deviation from that mean,

$$AR_{i,t} = R_{i,t} - \hat{\mu}_i, \quad \hat{\mu}_i = \frac{1}{120} \sum_{s=-140}^{-21} R_{i,s}, \quad (5.3)$$

and the cumulative abnormal return aggregates these over the window,

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{i,t}. \quad (5.4)$$

Each event is dated by its announcement, the point at which the information reaches markets, and that trading day is day 0. When an announcement falls on a non-trading day, day 0 is the first trading day after it, so the strikes of 28 February 2026 enter on Monday 2 March 2026. Because daily prices react to many things at once, the longer the window over which abnormal returns are summed, the weaker the claim that the measured movement belongs to the event alone ([MacKinlay 1997](#); [Kothari and Warner 2007](#)). Inference therefore rests on the short windows. The symmetric $CAR(-1, +1)$ guards against information reaching prices the day before the announcement, and $CAR(0, +1)$ and $CAR(0, +5)$ capture the announcement reaction. Both tariff announcements, Liberation Day and the tariff pause, came in the U.S. afternoon, just after 4 p.m. Eastern time on 2 April 2025 ([Benguria and Saffie 2025](#)) and in the early afternoon of 9 April 2025. That is after the noon pricing of the H.10 dollar indices and late in the pricing day of the currency panel, so the day-0 return contains at most a first partial reaction and the full response appears on the following trading day. Capturing announcements that arrive around or after the close is the standard reason for the two-day window. The longer cumulations $CAR(0, +10)$ and $CAR(0, +20)$ are reported as robustness checks on the durability of the response rather than as independent identification, and a long horizon $CAR(0, +50)$, a persistence check, is estimated on $[-170, -51]$ so the

estimation window stays clear of this check's wider $[-50, +50]$ event window.

The significance of a single CAR is assessed with

$$t = \frac{\text{CAR}_i(t_1, t_2)}{\hat{\sigma}_i \sqrt{L}}, \quad L = t_2 - t_1 + 1, \quad (5.5)$$

where $\hat{\sigma}_i$ is the estimation-window standard deviation of $\text{AR}_{i,t}$. The statistic is compared with the standard normal distribution, and the tables mark significance at the 10, 5 and 1 percent levels with its critical values of 1.65, 1.96 and 2.58. The difference between two events' CARs is tested with

$$\Delta = \text{CAR}_A(t_1, t_2) - \text{CAR}_B(t_1, t_2), \quad \text{s.e.}(\Delta) = \sqrt{L(\hat{\sigma}_A^2 + \hat{\sigma}_B^2)}, \quad (5.6)$$

where $\text{s.e.}(\Delta)$ is the standard error of the difference, the two-sample analogue of the CAR variance in [Campbell et al. \(1997\)](#). The two events are treated as independent because their windows do not overlap.

The constant-mean model is preferred to the market model because the object of study is the dollar itself. The choice matters in currency applications. [Kwok and Brooks \(1990\)](#), comparing competing abnormal-return specifications in foreign-exchange markets, find inference sensitive both to the currency in which returns are measured and to the benchmark model, so event-study practice developed for equities does not carry over automatically. When an event moves many currencies at once, a market model subtracts the common movement, the very thing this study wants to measure. Kwok and Brooks therefore prefer mean-adjusted returns when the whole unexpected movement is the quantity being measured. The natural market return for the currency baskets is the equal-weighted dollar basket, the average of the panel itself. Regressing dollar exchange rates on it would absorb the common dollar movement into the market term and leave little in the abnormal return. The market model is nevertheless reported as a robustness check,

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}, \quad \text{AR}_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}), \quad (5.7)$$

where $R_{m,t}$ is the equal-weighted dollar basket. The basket-level results in the Appendix leave the safe-haven ordering under both treatment shocks unchanged. When many correlated securities share a common event date, cross-sectional dependence inflates the test statistics, and the adjusted test of [Kolari and Pynnönen \(2010\)](#) corrects for it. Neither condition applies here. Inference is drawn at the basket level around single, non-overlapping event dates, so the plain t -test is kept.

5.3 Regression Specification

The event study measures whether the dollar moved. The regression measures whether its relationship with risk changed. The classical safe-haven property is that the dollar appreciates when global risk rises (Habib and Stracca 2012), so the daily broad-dollar return is regressed on a daily risk shock ΔR_t , interacted with a dummy for each crisis window, following the interaction tests of Baur and Lucey (2010),

$$r_t^{\text{USD}} = a + b_1 \Delta R_t + \sum_k \beta_k (\Delta R_t \times D_t^k) + \sum_k \gamma_k D_t^k + \varepsilon_t, \quad (5.8)$$

where the sum runs over the three crisis windows, $k \in \{\text{tariff, Hormuz, Ukraine}\}$. The dummy D_t^k marks the window of shock k . It is one on the event day and the twenty trading days after it, and zero on all other days. The coefficient b_1 gives the dollar's response to a risk shock on days outside these windows. The β_k measure how that response changes inside each crisis window, and the γ_k capture the dollar's average drift within each window, so a window in which the dollar simply trended does not show up as a changed risk response. The event dates and the twenty-day horizon are the same as in the event study, so both methods work with the same definition of each episode.⁵ The equation is estimated by ordinary least squares (OLS) with Newey–West standard errors with five lags (Newey and West 1987), which are heteroskedasticity- and autocorrelation-consistent (HAC) and therefore robust to the volatility clustering of daily returns. All computations use Python (version 3.14) with the pandas, statsmodels and matplotlib libraries, and the code and data are provided together with the thesis.

The choice of ΔR_t matters because no single index captures both shock types equally well. The VIX responds strongly to the trade-policy shock but only mildly to the Gulf escalation, and the news-based GPR index does the opposite and barely registers a tariff announcement. The regression is therefore estimated in four versions. The baseline uses the change in the VIX. The second adds the GPR index alongside the VIX, each with its own interaction terms, so stock-market risk and geopolitical risk enter separately. Their daily changes are almost uncorrelated, $\rho = -0.04$, so the data can tell the two effects apart. The third replaces both with the VXY Global index of implied U.S.-dollar exchange-rate volatility, a measure taken from the

⁵An alternative would be a DCC-GARCH model (dynamic conditional correlation with generalised autoregressive conditional heteroskedasticity), in which the correlation between the dollar and risk varies continuously over time (Engle 2002). The dummy-interaction design is kept because it is simpler and ties each hypothesis to a single coefficient.

5 Data and Methodology

currency market itself. The fourth combines the two news-based indices, GPR and TPU. Each risk shock is scaled by its own standard deviation over the full sample, so the coefficients are comparable across versions and read as the percent dollar response to a one-standard-deviation rise in risk. Results are reported in Section 6.6.

6 Empirical Results

This chapter presents the thesis’s empirical results. An event study, set out in detail in the next five sections, measures how the U.S. dollar, the cross-section of currency baskets, crude oil, gold and equities responded to the four shocks, the Liberation Day tariffs of April 2025, the twelve-day war of June 2025, the Hormuz crisis of 2026 and the Russian invasion of Ukraine in 2022. The regression in Section 6.6 then tests whether the relationship with risk that makes the dollar a safe haven held or broke across the episodes. Its results are read as supporting evidence for the event study rather than as an independent test.

Three of the four shocks are U.S.-linked. The Liberation Day tariff announcement is U.S.-originated, and the twelve-day war and the Hormuz crisis are military shocks in which the United States is involved as a co-belligerent. The Russian invasion of Ukraine is an external military/oil-supply shock that the United States did not author, and thus serves as the benchmark for the “classical” safe-haven response.

Abnormal returns are computed with the constant-mean model of Section 5.2. The normal return of each series is its mean over the estimation window of $[-140, -21]$ trading days, the abnormal return is the deviation from that mean, and the cumulative abnormal return $CAR(t_1, t_2)$ sums the abnormal returns over the event window from day t_1 to day t_2 . Conclusions are drawn from the short windows $CAR(-1, +1)$, $CAR(0, +1)$ and $CAR(0, +5)$, which capture the reaction to the announcement itself. Over longer windows other events affect returns as well, so the ten-, twenty- and fifty-day CARs are used to trace how long a response lasts, not to establish that there was one.¹ Significance is assessed with the t -test of Section 5.2. Tables 6.1 and 6.2 collect the main results, and the discussion below reads them event by event. The figures plot Brent, the global benchmark, and WTI is reported in Table 6.2.

6.1 Dollar Response to the Tariff Shock

The tariff announcement of 2 April 2025 triggered an immediate depreciation of the dollar. The broad dollar index fell 1.44% over the two-day announcement window and 1.70% over the symmetric $(-1, +1)$ window, both significant at the 1% level (Table 6.1, Panel A). The five-day CAR of -0.10% is close to zero because the dollar

¹The constant-mean specification is preferred to the market model because the natural market return for the currency baskets is the equal-weighted dollar basket itself, which is endogenous to the response under study (Section 5.2). Market-model results at the basket level are reported in the Appendix.

6 Empirical Results

recovered on 4 and 7 April, undoing the initial fall within the window. Day +5 falls on 9 April 2025, the day of the ninety-day tariff pause, so the five-day window ends on a second policy event. $CAR(-1, +1)$ and $CAR(0, +1)$ end on 3 April, four trading days before the pause, which makes them the cleaner measures of the announcement effect. The pause is dated as a sub-event with 9 April as day 0. Over its two-day window, 9 and 10 April, the dollar falls again, by 1.48%, and the fall continues, to -3.07% after five trading days and -4.51% after twenty. Measured from the announcement, the move persists rather than reverses over the longer horizons, -3.00% by ten trading days, -3.98% by twenty and -6.34% by fifty (Section 6.5), which is consistent with markets reading the tariffs as a lasting change in U.S. trade policy rather than a one-off shock.

U.S. markets responded with a broad flight from risk. Over the two-day announcement window, 2 and 3 April, the S&P 500 fell 4.33% and the VIX rose 8 points, both significant at the 1% level, while the 10-year Treasury yield fell about 12 basis points, a move that is not statistically significant. Equities sold off, Treasuries did not, and the dollar fell alongside equities rather than rising against them. Around the pause Treasuries sold off as well, the 10-year yield rising 24 basis points over the three days from 8 to 10 April, significant at the 5% level, while the dollar kept falling. Over 2 and 3 April the yen, Swiss franc and euro rose against the dollar by 2.48%, 2.89% and 2.44% respectively. Together they form the literature-standard safe basket, which rose 2.60%. The broader carry-sorted safe basket, which also contains low-yield Asian currencies, rose 1.52%. All of these moves are significant at the 1% level. Over five days the gains shrink to 2.12% for the literature-standard safe basket and 1.07% for the carry-sorted basket, because the dollar's recovery on 4 and 7 April pulled both back (Table 6.2). These five-day gains are significant only at the 10% level for the literature-standard safe basket, while not significant for the carry-sorted basket.

The oil-exporter and oil-importer baskets point the same way. Crude fell sharply on the announcement (Section 6.4), and over ten days the importer basket gained 2.84% while the exporter basket was essentially flat at +0.04%, a gap of 2.80 percentage points, significant at the 1% level (Figure 6.1). The gap does not isolate the oil channel, however. Three of the six importer currencies, the yen, baht and Taiwan dollar, also belong to the carry-sorted safe basket and had already risen as safe currencies. Taken together, these results match what the framework calls the erosion channel. A U.S.-originated shock weakened the dollar, and the demand for safety went to the yen, franc and euro instead, while oil and equities fell.

6 Empirical Results

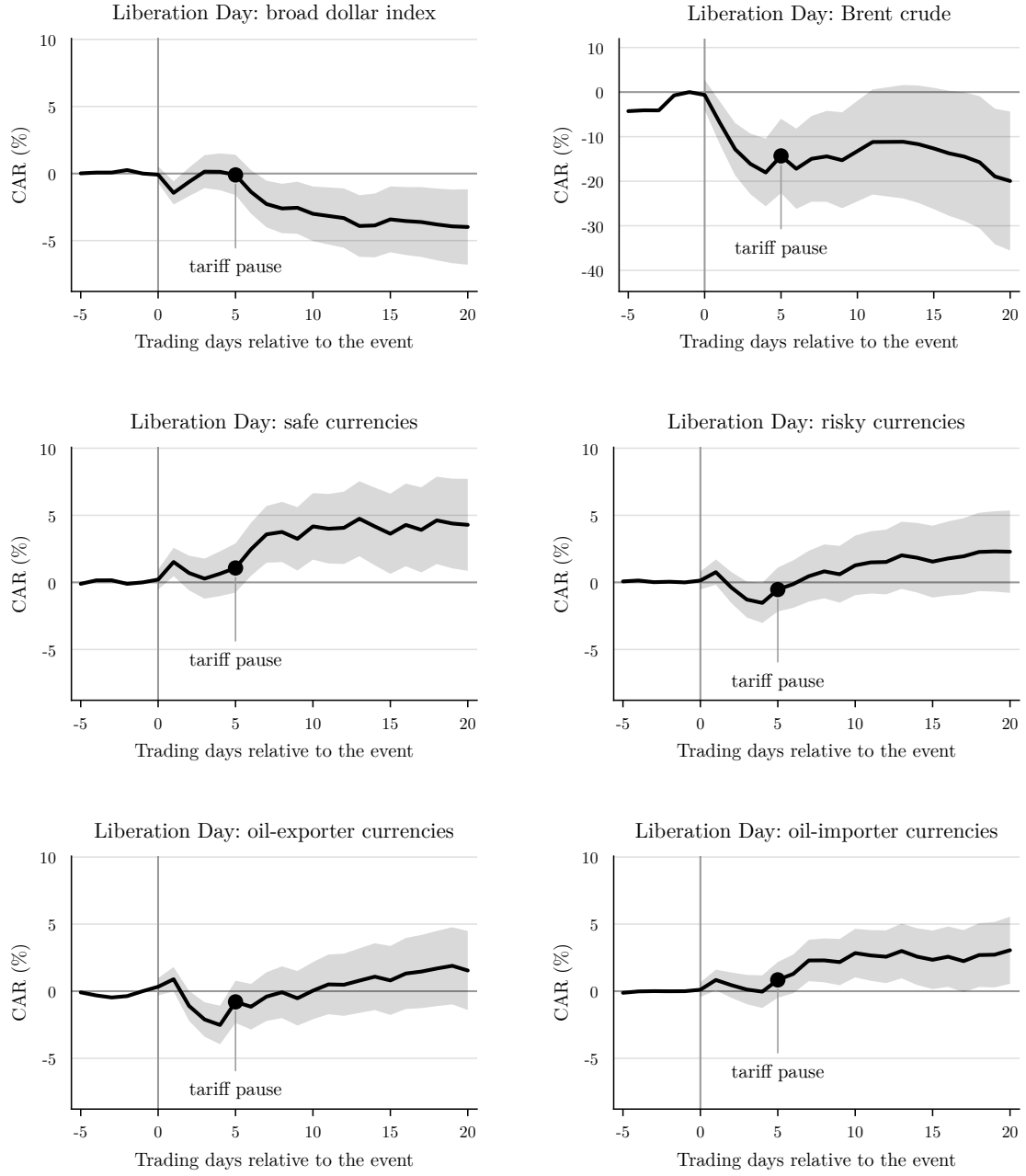


Figure 6.1: Liberation Day (2 April 2025): cumulative abnormal returns over -5 to $+20$ trading days, constructed as in Section 5.2. In each portfolio figure the dollar-index and currency-basket panels share one vertical scale, chosen for that figure, and Brent crude has its own. Every path is re-anchored to zero at day -1 so that the movement from day 0 onwards is the event response. The shaded band is ± 1.96 standard errors of the cumulative abnormal return as defined in Section 5.2. Where it clears zero, $CAR(0, h)$ is significant at the 5% level. Dots mark dated sub-events.

6 Empirical Results

Table 6.1: The U.S. dollar's response by event and horizon

Event	Headline windows			Longer horizons		
	(−1, +1)	(0, +1)	(0, +5)	(0, +10)	(0, +20)	(0, +50)
<i>Panel A. U.S.-originated trade-policy shock</i>						
Liberation Day	−1.70***	−1.44***	−0.10	−3.00***	−3.98***	−6.34***
Tariff pause	−1.48**	−1.48***	−3.07***	−3.44***	−4.51***	−6.74***
<i>Panel B. U.S.-involved military/oil-supply shocks</i>						
Twelve-day war	−0.28	+0.04	+0.88	−0.13	+0.35	−0.12
Hormuz crisis	+1.34***	+1.39***	+1.53***	+2.10***	+3.25***	−0.13
Hormuz ceasefire	−1.33***	−1.14***	−1.72***	−1.65**	−1.47	−0.04
<i>Panel C. External control</i>						
Ukraine invasion	+0.45	+0.49	+1.03*	+1.65**	+0.84	+3.12**

Notes. Cumulative abnormal returns of the broad nominal U.S. dollar index, in percent, constant-mean model. Estimation windows are as in Section 5.2, with the fifty-day column estimated on $[-170, -51]$. The three headline windows carry the conclusions. The longer horizons trace how long a response lasts (Section 6.5). Positive values denote dollar appreciation. The tariff pause and the Hormuz ceasefire are sub-events within their parent windows. The ceasefire's estimation window ends on 9 March and contains the first six trading days of the crisis, so its cells are read with corresponding caution. The May 2026 U.S. strikes and the June 2026 memorandum are not analysed (Section 6.5). *, **, *** denote significance at the 10%, 5% and 1% level.

6.2 Dollar Response to the Military Shocks

The dollar responded very differently to the two military shocks. It showed no significant move in the twelve-day war and rose as a classical safe haven in the Hormuz crisis. The twelve-day war disrupted oil supply only briefly, and the United States joined it six trading days after Israel's opening strike. The Hormuz crisis disrupted oil supply for months, and the United States was a belligerent from the first day.

The twelve-day war of June 2025 left the dollar essentially unmoved. The broad dollar index moved +0.04% over the two-day window and +0.88% over five days, neither significant (Table 6.1, Panel B), while Brent rose 11.25% over the same five days, significant at the 1% level (Figure 6.2). The U.S. strikes of 22 June 2025 enter the panel on day +6, so the three headline windows cover an opening week in which the war was fought between Israel and Iran alone, and only the ten- and twenty-day windows contain the U.S. strikes and the ceasefire of 24 June. The dollar moved significantly in neither stretch. Up to day +6 its cumulative abnormal return drifted

6 Empirical Results

to +1.01%, the same sign as the Ukraine benchmark but never significant, and by day +9 the gain was gone. An offset between the flight-to-the-dollar reflex and the erosion pressure once the United States took part would fit that reversal, but the ceasefire on the next trading day fits it equally well, so the episode identifies neither. The response is read as descriptive, and Section 6.5 adds that its benchmark is distorted by the Liberation Day crash, which leaves even its sign uncertain.

In the Hormuz crisis the dollar behaved as a classical safe haven. The broad index rose 1.39% over the two-day window and 1.53% over five days, both significant at the 1% level, and kept rising to +2.10% at ten days and +3.25% at twenty (Table 6.1, Panel B). Crude rose sharply from the first day (Section 6.4), but unlike under the tariff shock the oil price did not sort the currencies. Both oil baskets fell against the rising dollar over five days, the importers by 1.03% and the exporters by 0.69%, and the exporter-minus-importer spread was not significant at any horizon, +0.54 percentage points at twenty days (Figure 6.3). The sign of the gap fits higher crude cushioning the exporters, but the difference never became significant. The dollar rose even though the United States was a belligerent from the first day, so for a military/oil-supply shock the safe-haven reflex survived U.S. involvement, in contrast to the trade-policy shock. U.S. assets show the same contrast. The S&P 500 was little changed over the first week, and the 10-year Treasury yield rose 27 basis points by ten days and 39 by twenty, both significant at the 5% level, so yields moved with oil rather than with a flight into Treasuries while the dollar appreciated.

The Hormuz appreciation was temporary. By fifty days the broad-index CAR had returned to an insignificant -0.13% , so most of the move unwound. The clearest evidence that this was a genuine but temporary risk premium comes from the ceasefire of 7 April 2026, a sub-event within the crisis marked at day +26 in Figure 6.5. On the two-week ceasefire announcement, which came with an Iranian declaration that the Strait was open again to coordinated traffic, the dollar gave back 1.14% over the two-day window and 1.72% over five days, both significant at the 1% level, and Brent fell 21.58% within ten days. The premium that the threat put into the dollar came back out on de-escalation, which ties the exchange-rate move to the geopolitical risk it priced. The ceasefire's estimation window ends on 9 March and therefore contains the first six trading days of the crisis, so its numbers are read with the caution noted in Table 6.1.

6 Empirical Results

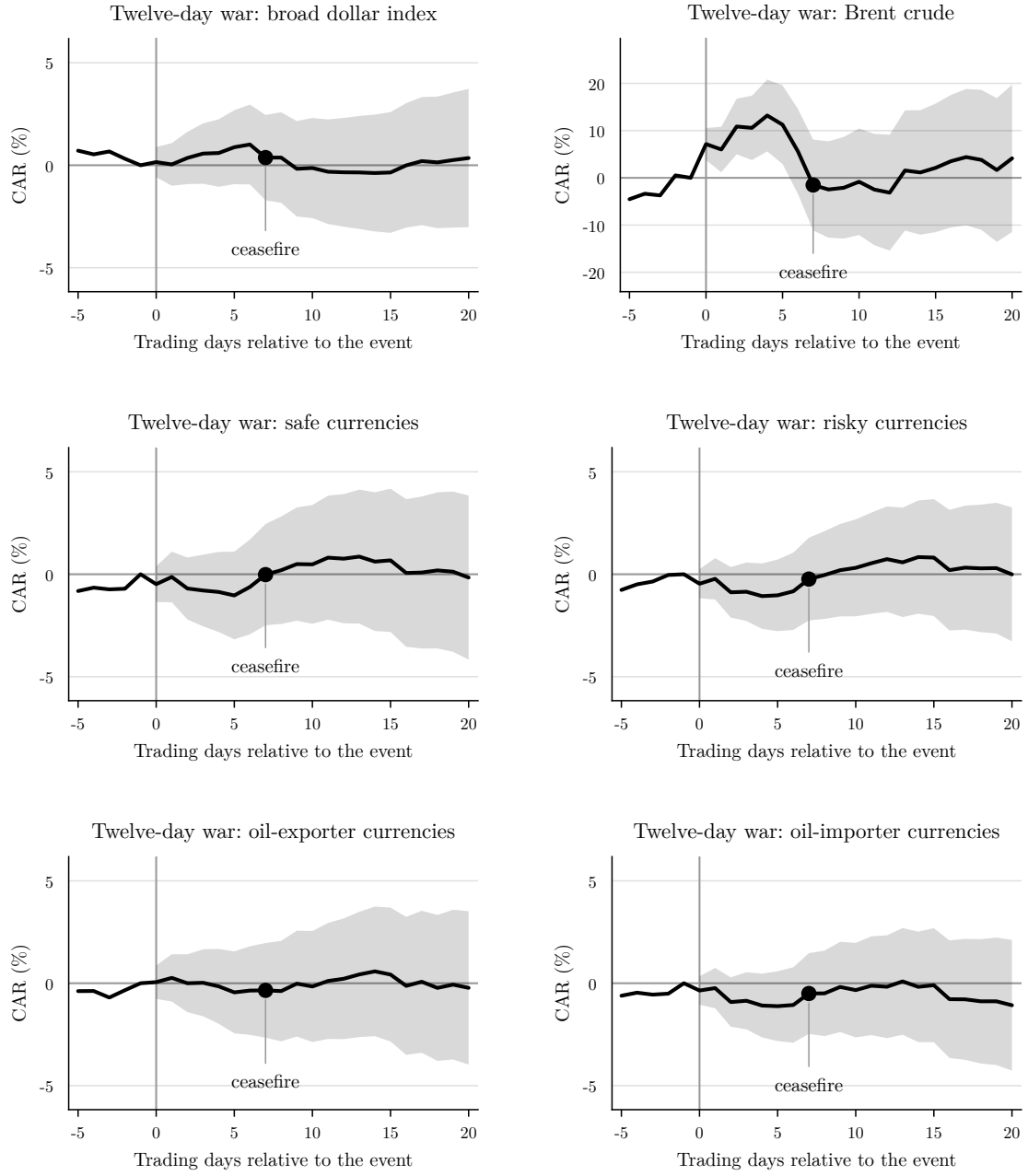


Figure 6.2: Twelve-day war (13 June 2025): cumulative abnormal returns over -5 to $+20$ trading days. Construction as in Figure 6.1. The 24 June ceasefire is marked.

6 Empirical Results

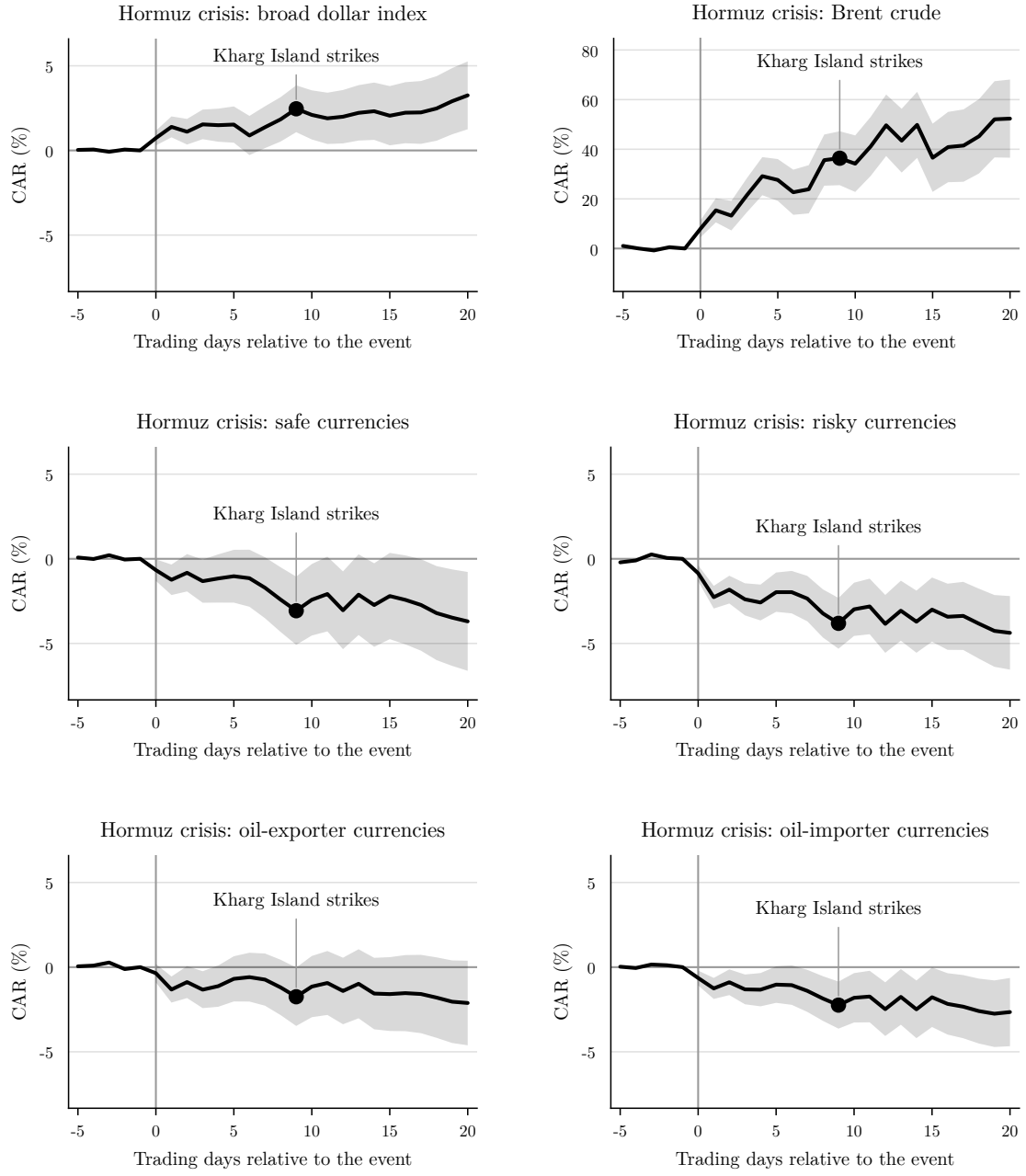


Figure 6.3: Hormuz crisis (28 February 2026): cumulative abnormal returns over -5 to $+20$ trading days. Construction as in Figure 6.1. The 13 March Kharg Island strikes are marked.

6 Empirical Results

Table 6.2: Cross-section of five-day responses: the mirror image

Series, CAR(0, 5) (%)	U.S.-originated	U.S.-involved		Control
	Liberation Day	Hormuz	Twelve-day war	Ukraine
Broad USD index	−0.10	+1.53***	+0.88	+1.03*
Dollar (equal-weighted)	+0.21	+1.36**	+0.83	+1.01
Safe currencies (carry)	+1.07	−1.03	−1.04	−1.07*
Risky currencies (carry)	−0.53	−1.97***	−1.03	−2.56*
Oil exporters	−0.80	−0.69	−0.45	+0.30
Oil importers	+0.84	−1.03*	−1.12	−0.74
Gold	−1.54	−4.54	−1.53	+1.25
S&P 500	−3.30*	−1.58	−1.22	+3.32
Euro Stoxx 50	−14.58***	−8.11***	−3.05	−5.96**
Brent crude	−14.32***	+27.63***	+11.25***	+13.82***
WTI crude	−13.14***	+34.55***	+10.14**	+14.38***
<i>Robustness: literature-standard baskets</i>				
Safe currencies (literature)	+2.12*	−1.23	−1.33	−0.69
Risky currencies (literature)	−1.69*	−1.18	−0.36	+0.24
<i>Robustness: euro-denominated commodities</i>				
Brent crude in EUR	−15.99***	+29.21***	+12.09**	+15.79***
Gold in EUR	−3.21	−2.97	−0.70	+3.23*

Notes. Five-day cumulative abnormal returns (%), constant-mean model, estimation window $[-140, -21]$. Positive values are appreciation against the dollar for the currency baskets and a rise in the index or price for all other series. *Safe/Risky (carry)* are the bottom and top forward-discount terciles (Section 5.1.3). The literature-standard baskets are listed in the Appendix. The bottom panel re-expresses the commodities in euros (Section 5.1.1). Gold's CARs are read for sign rather than magnitude (Section 6.4). *, **, *** denote significance at the 10%, 5% and 1% level.

6.3 Cross-Event Comparison

Reading across Tables 6.1 and 6.2, the two treatment shocks produce a mirror image. The trade-policy shock weakens the dollar and depresses oil, while the military/oil-supply shock strengthens the dollar and lifts oil. The dollar is therefore not a fixed safe haven or a fixed casualty. Its response is conditioned by the *type* of shock, even holding U.S. involvement fixed.

This contrast is not merely descriptive. Table 6.3 tests the difference in the dollar's CAR between events directly, and the separation is immediate: already over the first two trading days the broad dollar index is 2.83 percentage points lower under the tariff shock than under the Hormuz crisis ($t = -5.2$) and 1.93 points lower than under Ukraine ($t = -3.5$), both significant at the 1% level. The five-day difference narrows to -1.63 points ($t = -1.7$) as the dollar's recovery of 4 and 7 April, just

6 Empirical Results

before the pause, temporarily retraces its fall, before the gap re-opens to 5.10 points at ten days ($t = -4.0$) and 7.24 points at twenty ($t = -4.1$). Against Ukraine the pattern is the same (-4.65 at ten days, $t = -3.6$). The Hormuz and Ukraine responses, by contrast, share the same sign throughout: over the first two days the Hormuz bid is somewhat stronger ($+0.91$ points, $t = 2.0$), a difference of intensity rather than direction, at five and ten days the two are statistically indistinguishable, and at twenty days a marginal gap re-opens ($+2.41$, $t = 1.7$). The mirror image is thus driven by the trade-policy shock, the only episode in which the dollar moves *against* the classical safe-haven direction, and it does so identifiably from the first day.

Table 6.3: Difference in dollar CARs across events (broad USD index, %)

Contrast (A – B)	$\Delta\text{CAR}(0, h)$, percentage points			
	$h=1$	$h=5$	$h=10$	$h=20$
Liberation Day – Hormuz crisis	-2.83^{***}	-1.63^*	-5.10^{***}	-7.24^{***}
Liberation Day – Ukraine	-1.93^{***}	-1.14	-4.65^{***}	-4.83^{***}
Hormuz crisis – Ukraine	$+0.91^{**}$	$+0.50$	$+0.45$	$+2.41^*$
Liberation Day – Twelve-day war	-1.48^{**}	-0.98	-2.87^*	-4.33^*

Notes. Difference between two events' cumulative abnormal returns of the broad nominal U.S. dollar index (%), constant-mean model, estimation window $[-140, -21]$. Standard errors follow equation (5.6). Each window includes day 0 through day h , and events are treated as independent because their windows do not overlap. A positive value means the dollar rose more (or fell less) under shock A than shock B. $^{**}/^{***}$: 10/5/1%.

The Ukraine benchmark disciplines the interpretation. As an external military/oil shock, it generates the textbook response. The dollar appreciates ($+1.65\%$ at ten days, $+3.12\%$ at fifty, Table 6.1) and crude rises 13.82% (Table 6.2). As just shown, that response shares its sign throughout the first twenty trading days and, at five and ten days, its magnitude with Hormuz. Two features nonetheless distinguish the external from the U.S.-involved military shock. First, the Ukraine appreciation is durable whereas the Hormuz appreciation reverts, consistent with an external shock conferring a lasting safe-haven bid and a U.S.-involved one only a temporary one. Second, and most importantly for the thesis, the *trade-policy* shock breaks the pattern entirely: the only episode in which the dollar depreciates is the one the United States itself authored on the economic-policy front. The safe-haven function is intact for geopolitical/oil-supply shocks, including U.S.-involved ones, but it inverts for a U.S. trade-policy shock.

6 Empirical Results

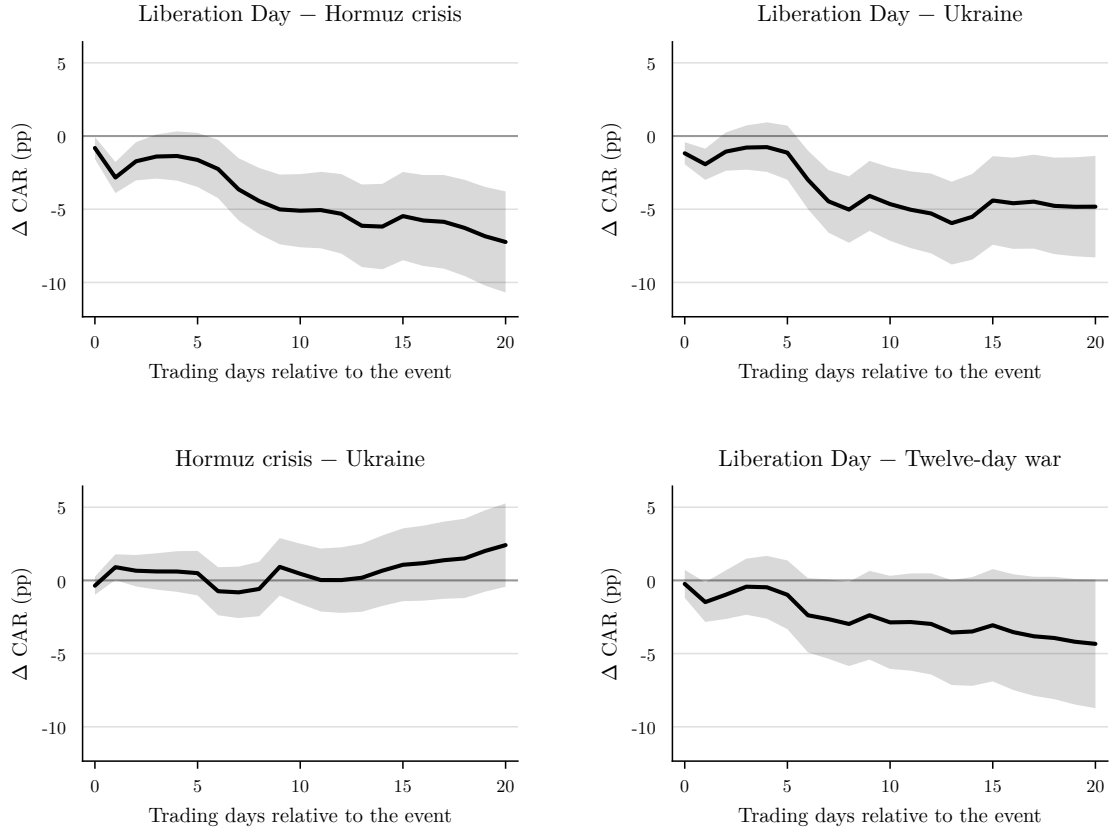


Figure 6.4: Difference in the broad dollar index's cumulative abnormal return between events, in percentage points over the first twenty trading days. The differences and their standard errors follow equation (5.6). The four contrasts are those of Table 6.3. All four panels share one vertical scale. The shaded band is ± 1.96 standard errors of the difference. Where it clears zero, the difference is significant at the 5% level. Differences start at day 0.

6.4 Oil, Gold and Equities

The commodity dimension reinforces the demand-versus-supply reading. Brent and WTI move in opposite directions across the two shock types, down $\sim 13\text{--}14\%$ on the tariff shock (a demand shock) and up $\sim 28\text{--}35\%$ on the Hormuz supply shock. The two benchmarks agree in sign and broad magnitude, so the result does not depend on the choice of crude reference. Nor does it depend on the currency of denomination: re-expressed in euros, Brent falls 15.99% on the tariff shock and rises 29.21% on Hormuz (Table 6.2), so the oil moves are genuine commodity repricings, not the dollar moving in the denominator (Section 6.5 returns to this). The oil

6 Empirical Results

response is also the channel behind the one clean cross-sectional sort in Table 6.2, the underperformance of the oil exporters when the tariff shock depresses crude, while under Hormuz the two oil baskets fall together and the spread stays insignificant.

Gold tells a subtler story than the “universal haven” a first intuition would suggest. Rather than rising around every shock, gold tracks the *inverse of the dollar* across the two treatment shocks: it gains when the tariff shock weakens the dollar (+6.07% at ten days, and +8.85% in the ten days from the tariff pause) and falls when the Hormuz shock strengthens it (−8.74% at ten days, deepening to −22.3% at twenty). The euro-denominated series sharpens this reading into a test: stripped of the dollar denominator, gold’s tariff-window gain essentially vanishes (+0.21% at ten days, insignificant, versus +6.07% in dollars), so most of gold’s “rally” around the tariff shock was the dollar falling underneath it. The one departure is the external control: around the Ukraine invasion gold rises *with* the dollar, and does so in euros as well (+6.71% at ten days, significant at the 1% level). This is a genuine dual-haven bid rather than a denomination effect. Gold is thus not an unconditional substitute for the dollar but a hedge against it. For the two treatment shocks it moves opposite the dollar, reinforcing that the dollar’s own conditional response, and not a generic flight to safety, is the organising variable of this chapter. Only for the external shock do both havens bid at once. One caveat tempers the magnitudes. Gold roughly tripled over the sample and its price peaked in late January 2026, days before the Hormuz estimation window closes, so the constant-mean baseline is unusually sensitive and the long-horizon gold CARs (e.g. Hormuz at twenty days) overstate the move. The *sign* is the robust feature, not the level.

Equities corroborate the risk-off reading without the dollar’s conditionality. The Euro Stoxx 50 falls around every shock over the five-day window, by 14.58% on Liberation Day, 8.11% around Hormuz and 5.96% after the Ukraine invasion, all significant at the 5% level or better, and by an insignificant 3.05% around the twelve-day war, so European risk assets sell off whether the shock is a trade war, an oil-supply scare, or an external invasion. The contrast with the dollar is the point: European equities are an unconditional casualty of geopolitical and policy shocks, whereas the dollar’s sign flips with the type of shock. The safe-haven question is therefore specific to the dollar, not a generic property of risk assets.

6.5 Robustness

Beyond the literature-standard basket panel of Table 6.2 and the interaction regression of Section 6.6, four further checks support the readings above. First, *longer windows*. Because many things move daily prices at once, the ten- and twenty-day cumulations are not treated as independent identification of the event effect. Read as durability checks, they confirm that both treatment shocks' headline signs extend rather than reverse (-3.00% and -3.98% for the tariff shock, $+2.10\%$ and $+3.25\%$ for Hormuz), and the fifty-day horizon (Table 6.1) then separates a level re-rating of the dollar (tariffs, deepening to -6.34%) from a transitory risk premium (Hormuz, largely reverting: -0.13% on the fifty-day estimation baseline, $+1.07\%$ when read on the same $[-140, -21]$ baseline as the shorter horizons, with about two thirds of the twenty-day peak given back). The +50-day figures (e.g. Figure 6.5) plot the dollar, crude and the two oil-exposure baskets in separate panels, the three currency panels on one scale and crude on its own, and mark neighbouring sub-events, making both the persistence and the ceasefire reversal visible. Second, *Brent versus WTI*: the crude responses share their sign and approximate size in every event, with WTI overshooting Brent materially only in the Hormuz spike (Table 6.2), so conclusions are invariant to the benchmark. Third, *denomination*: re-expressing the USD-priced commodities in euros (Table 6.2, bottom panel) leaves the crude responses essentially unchanged while gold keeps its sign but not its size. The former are genuine commodity repricings, while gold's dollar-hedge reading rests on the ten-day evidence of Section 6.4. Fourth, the market model with the equal-weighted dollar basket as market return, reported for the currency baskets in the Appendix, leaves the safe-haven ordering under both treatment shocks unchanged. It also sharpens the reading. Once the common dollar movement is removed, the safe basket beats the risky basket on Liberation Day ($+1.31\%$ against -0.33% over five days) and under Hormuz ($+0.67\%$ against -0.73%) alike, so the risk-off sorting among currencies is unconditional and what flips with the type of shock is whether the dollar takes part in it.

A final check concerns inference rather than the point estimates. The standard errors reported here follow the constant-mean convention of [MacKinlay \(1997\)](#): the residual variance is estimated on the pre-event window and the sampling error of the estimated normal return $\hat{\mu}$ is left out, so

$$\text{Var}(\text{CAR}) = L\hat{\sigma}^2. \quad (6.1)$$

6 Empirical Results

Two departures from that convention both widen the interval. Carrying the estimation error of $\hat{\mu}$ through gives

$$\text{Var}(\text{CAR}) = L\hat{\sigma}^2(1 + L/T), \quad (6.2)$$

with $T = 120$ estimation days, an inflation of the standard error of 0.8% at $L = 2$, 8.4% at $L = 21$ and 19.4% at $L = 51$. And the event windows are visibly more volatile than the estimation windows that benchmark them: for the broad dollar the event-window standard deviation is 1.99 times its estimation-window value around Ukraine, 1.72 around Liberation Day and 1.65 around Hormuz, and trimming the two largest days of each window still leaves a ratio of 1.3 to 1.5, so this is sustained volatility rather than the announcement jump alone. A variance estimated before the event therefore understates the variance during it. Table 6.4 recomputes the headline contrast, Liberation Day – Hormuz at twenty days, under each departure and under both. The point estimate never moves and the contrast rejects at the 5% level under every treatment. What the corrections cost is the 1% starring, not the finding. Since the same widening applies to every cell, the stars printed at the ten- and twenty-day horizons are read as an upper bound on the strength of the evidence rather than as its measure, which is why inference rests on the short windows (Section 5.2).

Table 6.4: Sensitivity of the headline cross-event contrast to the variance estimate

Variance used in the standard error	ΔCAR (pp)	s.e. (pp)	t	Rejects at
As reported: estimation-window $\hat{\sigma}$, $L\hat{\sigma}^2$	−7.24	1.76	−4.11	1%
+ estimation error in $\hat{\mu}$: $L\hat{\sigma}^2(1 + L/T)$	−7.24	1.91	−3.79	1%
Event-window $\hat{\sigma}$ instead	−7.24	2.99	−2.42	5%
Both corrections	−7.24	3.24	−2.23	5%

Notes. Liberation Day – Hormuz crisis, broad nominal U.S. dollar index, $\text{CAR}(0, +20)$, constant-mean model, the headline cell of Table 6.3. $L = 21$, $T = 120$. The first row is the specification reported throughout, following [MacKinlay \(1997\)](#). The second adds the sampling error of the estimated normal return. The third replaces the pre-event residual standard deviation with the one measured inside the event window. The point estimate is identical in every row. Only the standard error moves.

Two limitations bound the long-horizon results. The ± 50 -day windows of the two 2025 events overlap, since the twelve-day war begins 52 trading days after Liberation Day and its own window reaches back into Liberation Day’s tail, so their long-horizon CARs are not fully independent. The persistence figures are therefore read event by event, while the formal cross-event *difference* tests (Table 6.3) use horizons from one to twenty days, at which the windows of the compared pairs do not overlap. A related caveat applies to the twelve-day war itself. Its estimation window contains

6 Empirical Results

the Liberation Day crash, which distorts the benchmark in both moments: the 41 trading days of the Liberation Day event window drag the estimated mean down (from +0.014 to -0.027 percent per day), mechanically raising the war's measured abnormal returns. On a benchmark estimated without those days its twenty-day dollar CAR is -0.50 rather than $+0.35$ percent. Those days also inflate the estimated standard deviation, biasing the tests toward insignificance. The war's muted response is therefore read as descriptive context rather than as a positive finding, with sign as well as significance uncertain. And two further Gulf events are not analysed. The estimation window of the May 2026 U.S. strikes runs from November 2025 to late April 2026 and therefore covers the Hormuz crisis itself, so their abnormal returns would be measured against a benchmark that the episode under study has already inflated. The memorandum of understanding of 17 June 2026 falls too close to the sample end of 30 June 2026 to allow a full event window.

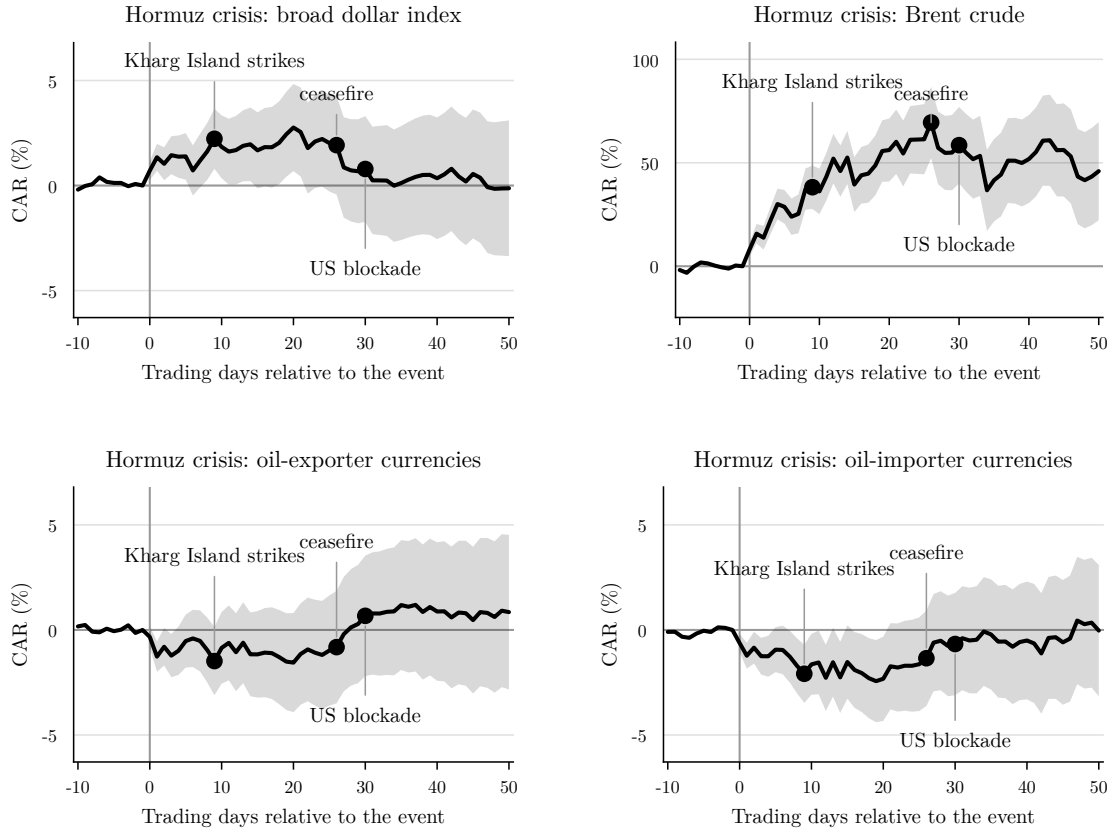


Figure 6.5: Hormuz crisis, -10 to $+50$ trading days, estimation window $[-170, -51]$ so that it does not overlap the wider event window. Construction as in Figure 6.1. The 13 March Kharg Island strikes, the 7 April two-week ceasefire and the 13 April naval blockade are marked.

6.6 Regression Results

The event study above shows *that* the dollar moved in opposite directions across the two shock types. This section asks the sharper question directly. Did the dollar’s defining safe-haven property, appreciating when global risk rises, hold or break in each episode? The dummy-interaction regression specified in Section 5.3 answers it with ordinary least squares and Newey–West standard errors, estimated in four versions that differ only in the measure of risk.

Table 6.5: Did the dollar’s safe-haven reflex survive each shock? (OLS, HAC standard errors)

	(A) VIX	(B) VIX + GPR	(C) VXY	(D) GPR + TPU
Δ VIX (normal)	+0.069*** (+7.0)	+0.070*** (+7.1)	—	—
× Tariff	−0.071*** (−2.6)	−0.067*** (−2.7)	—	—
× Hormuz	+0.057 (+1.3)	+0.062 (+1.3)	—	—
× Ukraine	+0.165*** (+4.8)	+0.168*** (+4.0)	—	—
Δ GPR (normal)	—	+0.013** (+2.0)	—	+0.011* (+1.7)
× Tariff	—	+0.154* (+1.8)	—	+0.180* (+1.8)
× Hormuz	—	+0.002 (+0.0)	—	−0.011 (−0.3)
× Ukraine	—	−0.000 (−0.0)	—	−0.025 (−0.5)
Δ VXY (normal)	—	—	+0.107*** (+6.9)	—
× Tariff	—	—	−0.127*** (−3.2)	—
× Hormuz	—	—	+0.161*** (+3.9)	—
× Ukraine	—	—	+0.117*** (+4.3)	—
Δ TPU (normal)	—	—	—	−0.003 (−0.5)
× Tariff	—	—	—	−0.022 (−1.1)
× Hormuz	—	—	—	+0.051 (+0.5)
× Ukraine	—	—	—	+0.032 (+0.1)
Window dummies	Yes	Yes	Yes	Yes
N	1954	1954	1954	1954
R^2	0.057	0.061	0.137	0.009

Notes. OLS of the daily broad-dollar return (percent) on daily risk shocks, their interactions with crisis-window dummies, and the window (level) dummies. Newey–West (HAC, 5 lags) standard errors are used, with t -statistics in parentheses. Each risk shock is standardized to unit variance, so coefficients read as the percent dollar move on a one-standard-deviation risk-off day. Crisis windows run from the event day to +20 trading days. VIX: CBOE (FRED VIXCLS); GPR: [Caldara and Iacoviello \(2022\)](#) daily index; VXY: J.P. Morgan VXY Global FX implied volatility (LSEG .JPMVXYGL, mid); TPU: [Caldara et al. \(2020\)](#) daily index, current vintage (downloaded 23 July 2026). A positive slope is safe-haven behaviour (risk up $\rightarrow$ dollar up). A negative interaction weakens it. */**/***: 10/5/1%.

The three specifications using market-priced risk measures, A to C in Table 6.5, reach the same verdict, while the news-based specification D does not, as set out at the end of this section. On normal days the dollar is a safe haven. In the baseline

6 Empirical Results

VIX specification a one-standard-deviation risk-off day lifts the broad dollar by 0.069% ($t = 7.0$). During the tariff window this reflex disappears, the interaction is equal and opposite (-0.071 , $t = -2.6$, significant at the 1% level), and the within-window sensitivity is essentially zero. The dollar stopped behaving as a haven precisely when the United States was the source of the shock. During the Hormuz window the interaction is positive but insignificant ($+0.057$, $t = 1.3$), leaving the normal relationship intact, and during the Ukraine window the reflex strengthens ($+0.165$, $t = 4.8$). Controlling for geopolitical news changes none of this. In the joint specification the VIX terms are essentially unchanged, the GPR index enters with a small positive normal-day slope, and its interactions are at most marginally significant. The currency-market measure sharpens the result. Implied dollar exchange-rate volatility, the VXY index, explains daily dollar returns more than twice as well as the VIX ($R^2 = 0.137$ against 0.057), and each of its window interactions is significant with the sign the safe-haven reading predicts, positive on normal days ($+0.107$, $t = 6.9$), erased during the tariff window (interaction -0.127 , $t = -3.2$, so the within-window slope is again near zero), and reinforced during the Hormuz (interaction $+0.161$, $t = 3.9$) and Ukraine ($+0.117$, $t = 4.3$) windows. The one measure priced in the currency market itself thus registers both the breakdown and the survival of the reflex within a single specification. The news-count pair, by contrast, has little daily explanatory power ($R^2 = 0.009$), with only the GPR interaction in the tariff window marginally positive ($+0.180$, $t = 1.8$), an honest negative for daily news-based measures. The estimated window dummies, not tabulated, echo the event study, with the dollar's average drift negative in the tariff window and positive in the Hormuz window in all four specifications, and positive in the Ukraine window in three of the four.

The regression thus supports the event study's conclusion. In the three specifications using market-priced risk measures the dollar's safe-haven function broke down in the U.S. trade-policy shock and remained intact or strengthened in the military and oil-supply shocks. The news-based specification, which explains little of the daily variation, does not reproduce the tariff-window breakdown. Its GPR interaction is positive and significant only at the 10% level and its TPU interaction is insignificant, so the verdict rests on the specifications with market-priced measures.

7 Discussion

This chapter draws the event-study and regression results together, relates them to the literature, and sets out their implications and limits.

7.1 Findings in Light of the Literature

The results speak directly to the six literatures set out in Chapter 2. The sharpest verdict is for the *dollar-erosion* strand. [Kamin \(2025\)](#) and [Jiang et al. \(2026\)](#) read the 2025 tariff episode as evidence that the dollar has stopped behaving as a safe haven, and [Hassan et al. \(2025\)](#) supply the theory: a trade war that isolates U.S. goods markets erodes the currency’s safety premium. The findings here both confirm and bound that claim. They confirm it for the tariff shock, when the dollar depreciates and its risk sensitivity collapses to zero, but they bound it by showing that the erosion is not a general weakening: the very same dollar appreciates around the contemporaneous Hormuz shock. What the dollar-erosion papers identify as a regime change is, on this evidence, specific to U.S.-originated economic-policy shocks rather than a property of the post-2025 dollar as such.

For the *safe-haven* tradition ([Ranaldo and Söderlind 2010](#); [Habib and Stracca 2012](#); [De Bock and de Carvalho Filho 2015](#)) the evidence is corroborative. The dollar performs as that literature predicts around both military/oil-supply shocks, appreciating as risk rises, and the formal difference test cannot separate the U.S.-implicated Hormuz response from the external Ukraine benchmark, so the haven bid survives even U.S. involvement when the shock is a security one. The conditional, crisis-contingent definition of [Baur and Lucey \(2010\)](#) is thus vindicated for the dollar: its haven status is real but state-dependent. That same framing organises the *gold* result. Gold hedges against the dollar rather than serving as a universal safe haven. It bids when the tariff shock weakens the dollar and falls when the Hormuz shock strengthens it, rising alongside the dollar only for the external Ukraine shock. This is exactly the state-dependence Baur and Lucey’s distinction implies.

The *commodity-currency* literature ([Y.-C. Chen and Rogoff 2003](#); [Kilian 2009](#)) is borne out only in part. At the five- to twenty-day horizons the exporter–importer spread carries the sign of the crude move in the three main episodes, but the clearest separation occurs after the tariff shock, the Hormuz spread remains insignificant, and Ukraine provides weaker evidence only at twenty days (Section 7.3). Finally, the *geopolitical-risk* literature ([Caldara and Iacoviello 2022](#)) supplies both the measure-

ment and a mechanism. The within-crisis ceasefire reversal, which unwinds the dollar premium as the threat recedes, is the threat-versus-realisation distinction Caldara and Iacoviello emphasise, mapped onto the exchange rate.

7.2 The Dollar’s Safe Haven Is Conditional on Shock Type

The central finding is that the dollar’s safe-haven response is conditioned by the *type* of shock, not merely by who causes it. The dollar appreciated around both military/oil-supply shocks, the external Ukraine invasion and the U.S.-involved Hormuz crisis, and the formal difference test cannot distinguish the two responses at five and ten days, even though the United States was a belligerent in one and merely witnessed the other. Yet it depreciated around the one purely economic-policy shock it created, the Liberation Day tariffs. Origin alone does not explain the pattern. The nature of the shock does. This refines the “dollar-erosion” reading (Kamin 2025; Jiang et al. 2026): the safe-haven function is not weakening in general but is specifically suspended when the United States is itself the source of an economic-policy shock, while it survives intact, despite the subsequent reversion, for the security shocks that the safe-haven tradition has always emphasised (Ranaldo and Söderlind 2010; Habib and Stracca 2012).

The regression sharpens this further. Estimated over the full sample, the dollar’s sensitivity to risk is significantly positive on normal days, reflecting the textbook safe-haven reflex, but the interaction term shows it collapsing to essentially zero inside the tariff window, while it remains intact during the Hormuz and Ukraine windows. The breakdown is thus not merely a difference in the *level* of the dollar across episodes but a change in its very *relationship with risk*: during the U.S. trade-policy shock the dollar stopped trading like a safe haven, and only then.

Read against the hypotheses of Section 3.4, the evidence adjudicates all three. **H1 is supported**: the dollar appreciated around both military/oil-supply shocks, depreciated around the trade-policy shock, and the difference-in-CARs test separates the tariff episode from both others while leaving Hormuz and Ukraine indistinguishable at the five- and ten-day horizons, with only a difference of intensity on impact and a marginal gap at twenty days. **H2 receives mixed evidence rather than robust support**: in the U.S.-originated tariff episode gold’s ten-day cumulative abnormal return is positive in dollars (+6.07%, significant at the 5% level) but small and insignificant in euros (+0.21%), and its five-day dollar-denominated estimate is

negative and insignificant (-1.54%), so the prediction is met at the ten-day dollar-denominated horizon and the result is sensitive to horizon and denomination. Gold's negative cumulative abnormal returns in the Hormuz crisis (-22.3% at twenty days, significant at the 1% level) further weaken its interpretation as an origin-blind safe-haven benchmark (Section 7.3). **H3 is partly supported:** at the five- to twenty-day horizons the exporter–importer spread carries the sign of the crude response in the three main episodes, negative under the tariff shock and positive under Hormuz and Ukraine, but the separation is significant at the 1% level only for the tariff shock and, more weakly, at twenty days for Ukraine (Section 7.3).

7.3 Gold and the Oil Channel

Gold does not behave as a universal safe haven, an asset that is uncorrelated or negatively correlated with risky assets in times of market stress in the sense of [Baur and Lucey \(2010\)](#), whatever the source of that stress. Instead it tracks the inverse of the dollar across the two U.S.-linked shocks, rising when the tariff shock weakened the dollar, falling when the Hormuz shock strengthened it, and bids alongside the dollar only for the external Ukraine shock. Gold is thus better described as a hedge *against* the dollar than as a substitute for it, which reinforces, rather than competes with, the chapter's main message: the dollar's own conditional move is the organising variable, and gold prices it from the other side.

The oil channel sorts the currencies as the commodity-currency literature predicts only in part. The clearest exporter–importer separation came after the tariff announcement, when crude fell, with an exporter-minus-importer spread of -2.80 percentage points over days 0 to 10, significant at the 1% level. Under Hormuz, when crude surged, the spread was positive but insignificant at five, ten and twenty days, and around the Ukraine invasion it reached $+4.38$ percentage points at twenty days, significant only at the 10% level. The pattern is consistent with the finding of [H. Chen et al. \(2016\)](#) that dollar exchange rates respond to demand-driven rather than supply-driven oil moves, although their monthly structural model and this daily event study identify different objects, and the overlap between the oil-importer and safe baskets (Section 6.1) prevents the spread from isolating oil exposure from safe-haven characteristics. For currencies such as the yen, safe-haven demand and higher oil-import costs may exert opposing pressures during an oil-supply shock.

7.4 Policy Implications

Three implications follow. For reserve managers and central banks, the dollar’s safe-haven status cannot be treated as unconditional: hedging programmes calibrated on the assumption that the dollar always rallies in risk-off episodes will misfire precisely when the shock originates in U.S. economic policy. For portfolio managers, diversification should be thought of across *shock types*, not only across assets. Gold and competing-haven currencies, not the dollar, carried the flight to quality in the tariff episode. For the broader debate on the dollar’s reserve role ([Obstfeld and Zhou 2023](#); [Jiang et al. 2026](#)), the evidence is a caution against over-reading any single episode: the 2025 tariff shock did dent the dollar, but the 2026 security shock showed the haven bid still firmly in place.

7.5 Limitations

Several caveats bound the results. Two Gulf events, the U.S. strikes of 25 May 2026 and the memorandum of understanding of 17 June 2026, are not analysed. The estimation window of the first contains 40 of its 120 trading days on or after the 2 March onset of the Hormuz crisis, so its benchmark would be contaminated, and the second lies too close to the sample end of 30 June 2026 for the longer windows (Section 6.5). The regression’s crisis windows are defined around the events rather than by a structural break test, and the VIX–dollar link it measures is a reduced-form association, not a causal channel. Gold’s exceptional 2025–26 trend makes its constant-mean CARs sensitive in level, so they are read for sign rather than magnitude. As a single-country event study around a small number of recent, partly overlapping episodes, the design trades breadth for the clean U.S.-versus-external contrast, and without forward/basis data beyond the one-month tenor, covered-interest-parity deviations are left for future work.

8 Conclusion

This thesis asked whether the U.S. dollar’s safe-haven response depends on the type of shock that triggers a crisis, even when the United States is implicated in the shock itself. Two recent episodes provided an unusually clean test: the 2025 Liberation Day tariffs, a purely U.S. economic-policy shock, and the 2026 Iran/Hormuz crisis, a geopolitical oil-supply shock, with the 2022 Russian invasion of Ukraine as an external control.

The evidence gives a clear answer. The two shock types produced a mirror image. The dollar depreciated around the tariff shock by over three and a half percent on the broad index within twenty trading days, while it appreciated around the Hormuz crisis and around Ukraine, and a formal difference test confirms the tariff response is significantly distinct from both security episodes, which are themselves statistically indistinguishable from one another at the five- and ten-day horizons. The dollar’s haven status is therefore conditional on the *nature* of the shock, not on its origin: it survives a U.S.-involved security shock but inverts for a U.S.-involved economic-policy shock. A complementary regression supports the breakdown statistically: the dollar’s safe-haven sensitivity to risk, significantly positive on normal days, collapses to zero in the tariff window, while holding through the Hormuz and Ukraine episodes. Two further results round out the picture: gold acts as a hedge against the dollar rather than as a universal safe haven, and the clearest exporter–importer separation occurs after the tariff shock, when crude fell, while the Hormuz spread remains insignificant and Ukraine provides weaker evidence at twenty days.

The thesis contributes the first systematic side-by-side comparison of the Liberation Day and Iran/Hormuz episodes for currency markets, carries a gold and oil benchmark through the same design to connect the safe-haven and commodity-currency literatures, and extends standard cross-sectional currency tools to mid-2026 on a consistent LSEG currency panel.

Several extensions suggest themselves. As the geopolitical situation continues to evolve, the sample can be extended so that the most recent episodes acquire full post-event windows. Richer forward and basis data would allow covered-interest-parity deviations to be brought into the analysis, and intraday data would tighten the identification around the announcement windows. A dynamic conditional-correlation (DCC-GARCH) model of the dollar–VIX relationship would test the safe-haven-breakdown hypothesis directly and is the natural next layer. Finally, a formal model of shock-dependent safe-haven demand would give the empirical pattern documented

8 *Conclusion*

here a theoretical foundation. In such a model, the currency of the shock's originator would lose its haven premium for policy shocks but not for security shocks.

9 References

- Albulescu, C. T. and Ajmi, A. N. (2021): “Oil Price and US Dollar Exchange Rate: Change Detection of Bi-directional Causal Impact.” In: *Energy Economics* 100, p. 105385.
- Arslanalp, S., Eichengreen, B. and Simpson-Bell, C. (2022): “The Stealth Erosion of Dollar Dominance and the Rise of Nontraditional Reserve Currencies.” In: *Journal of International Economics* 138, p. 103656.
- Asness, C. S., Moskowitz, T. J. and Pedersen, L. H. (2013): “Value and Momentum Everywhere.” In: *The Journal of Finance* 68.3, pp. 929–985.
- Baur, D. G. and Lucey, B. M. (2010): “Is Gold a Hedge or a Safe Haven? An Analysis of Stocks, Bonds and Gold.” In: *Financial Review* 45.2, pp. 217–229.
- Benguria, F. and Saffie, F. E. (2025): *Rounding Up the Effect of Tariffs on Financial Markets: Evidence from April 2, 2025*. NBER Working Paper 34036. National Bureau of Economic Research.
- Caldara, D. and Iacoviello, M. (2022): “Measuring Geopolitical Risk.” In: *American Economic Review* 112.4, pp. 1194–1225.
- Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A. and Raffo, A. (2020): “The Economic Effects of Trade Policy Uncertainty.” In: *Journal of Monetary Economics* 109, pp. 38–59.
- Campbell, J. Y., Lo, A. W. and MacKinlay, A. C. (1997): *The Econometrics of Financial Markets*. Princeton, NJ: Princeton University Press.
- Cheema, M. A., Faff, R. and Szulczyk, K. R. (2022): “The 2008 Global Financial Crisis and COVID-19 Pandemic: How Safe Are the Safe Haven Assets?” In: *International Review of Financial Analysis* 83, p. 102316.
- Chen, Y.-C. and Rogoff, K. (2003): “Commodity Currencies.” In: *Journal of International Economics* 60.1, pp. 133–160.
- Chen, H., Liu, L., Wang, Y. and Zhu, Y. (2016): “Oil Price Shocks and U.S. Dollar Exchange Rates.” In: *Energy* 112, pp. 1036–1048.
- CNN (2026): *US Strikes on Iranian Missile Launch Sites and Boats, Iran Threatens to Retaliate*. Live coverage, 25–26 May 2026. URL: <https://www.cnn.com/2026/05/25/world/live-news/iran-war-us-peace-deal> (visited on 08/06/2026).
- Congressional Research Service (2025): *Israel-Iran Conflict, U.S. Strikes, and Cease-fire*. CRS In Focus IF13032. Updated 26 June 2025. Congressional Research Service. URL: <https://www.congress.gov/crs-product/IF13032> (visited on 08/06/2026).

- Congressional Research Service (2026a): *Presidential 2025 Tariff Actions: Timeline and Status*. CRS Report R48549. Updated 12 January 2026. Congressional Research Service. URL: <https://www.congress.gov/crs-product/R48549> (visited on 08/06/2026).
- (2026b): *U.S. Conflict with Iran*. CRS Report R48887. 26 March 2026. Congressional Research Service. URL: <https://www.congress.gov/crs-product/R48887> (visited on 08/06/2026).
 - (2026c): *U.S.-Iran Ceasefire and Negotiations: Assessment and Issues for Congress*. CRS Insight IN12678. Updated 13 May 2026. Congressional Research Service. URL: <https://www.congress.gov/crs-product/IN12678> (visited on 08/06/2026).
- Dalgic, H. C. and Ozhan, G. K. (2025): *Dominant Currency Pricing and Currency Risk Premia*. Discussion Paper 717. CRC TR 224.
- De Bock, R. and de Carvalho Filho, I. (2015): “The Behavior of Currencies during Risk-off Episodes.” In: *Journal of International Money and Finance* 53, pp. 218–234.
- Engle, R. (2002): “Dynamic Conditional Correlation: A Simple Class of Multivariate Generalized Autoregressive Conditional Heteroskedasticity Models.” In: *Journal of Business & Economic Statistics* 20.3, pp. 339–350.
- European Central Bank (Nov. 2025): *Financial Stability Review*. Report. European Central Bank.
- (2026): *Economic Bulletin, Issue 2/2026*. Report. European Central Bank.
- Fattouh, B. and Economou, A. (Apr. 2026): *The Anatomy of the Strait of Hormuz Oil Shock*. OIES Paper. Oxford Institute for Energy Studies.
- Gopinath, G., Boz, E., Casas, C., Díez, F. J., Gourinchas, P.-O. and Plagborg-Møller, M. (2020): “Dominant Currency Paradigm.” In: *American Economic Review* 110.3, pp. 677–719.
- Gourinchas, P.-O. and Rey, H. (2005): *From World Banker to World Venture Capitalist: US External Adjustment and the Exorbitant Privilege*. NBER Working Paper 11563. National Bureau of Economic Research.
- Gourinchas, P.-O., Rey, H. and Govillot, N. (2017): “Exorbitant Privilege and Exorbitant Duty.” In: Working paper, revised October 2017.
- Habib, M. M. and Stracca, L. (2012): “Getting Beyond Carry Trade: What Makes a Safe Haven Currency?” In: *Journal of International Economics* 87.1, pp. 50–64.
- Hassan, T. A., Mertens, T. M., Wang, J. and Zhang, T. (2025): *Trade War and the Dollar Anchor*. Conference Draft. Fall 2025. Brookings Papers on Economic Activity.

9 References

- House of Commons Library (2025): *Sanctions against Russia (February 2022 to January 2025)*. Research Briefing CBP-9481. 21 January 2025. House of Commons Library. URL: <https://commonslibrary.parliament.uk/research-briefings/cbp-9481/> (visited on 08/06/2026).
- (2026a): *Israel/US-Iran Conflict 2026: Background and UK Response*. Research Briefing CBP-10521. 24 April 2026. House of Commons Library. URL: <https://commonslibrary.parliament.uk/research-briefings/cbp-10521/> (visited on 08/06/2026).
 - (2026b): *US-Iran Ceasefire and Nuclear Talks in 2026*. Research Briefing CBP-10637. 2 July 2026. House of Commons Library. URL: <https://commonslibrary.parliament.uk/research-briefings/cbp-10637/> (visited on 08/06/2026).
- International Monetary Fund (Apr. 2025): *Global Financial Stability Report, Chapter 2: Geopolitical Risks: Implications for Asset Prices*. Report. International Monetary Fund.
- (Apr. 2026): *War in the Middle East: Economic Spillovers and Policy Challenges*. Regional Economic Outlook: Middle East and Central Asia. International Monetary Fund.
- Jiang, Z., Krishnamurthy, A. and Lustig, H. (2021): “Foreign Safe Asset Demand and the Dollar Exchange Rate.” In: *Journal of Finance* 76.3, pp. 1049–1089.
- Jiang, Z., Krishnamurthy, A., Lustig, H. and Richmond, R. J. (2026): “Dollar Erosion: Understanding the Loss of Reserve Currency Status.” In: Working paper.
- Kalemli-Özcan, Ş., Soylu, C. and Yildirim, M. A. (2026): *Global Trade, Tariff Uncertainty and the U.S. Dollar*. NBER Working Paper 34728. National Bureau of Economic Research.
- Kamin, S. B. (2025): *Dollar Movements and Dollar Dominance in the Aftermath of Liberation Day*. AEI Economics Working Paper 2025-06. American Enterprise Institute.
- Khalil, M. and Strobel, F. (2024): “US Trade Policy and the US Dollar.” In: *Journal of International Economics* 151. Article 103970.
- Kilian, L. (2009): “Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market.” In: *American Economic Review* 99.3, pp. 1053–1069.
- Kilian, L., Plante, M. D., Richter, A. W. and Zhou, X. (2026): *The Impact of the 2026 Iran War on U.S. Inflation: A Scenario Analysis*. Working Paper 2609. Federal Reserve Bank of Dallas.

- Kolari, J. W. and Pynnönen, S. (2010): “Event Study Testing with Cross-sectional Correlation of Abnormal Returns.” In: *Review of Financial Studies* 23.11, pp. 3996–4025.
- Kothari, S. P. and Warner, J. B. (2007): “Econometrics of Event Studies.” In: *Handbook of Empirical Corporate Finance*. Ed. by B. E. Eckbo. Vol. 1. Amsterdam: Elsevier. Chap. 1, pp. 3–36.
- Kwok, C. C. Y. and Brooks, L. D. (1990): “Examining Event Study Methodologies in Foreign Exchange Markets.” In: *Journal of International Business Studies* 21.2, pp. 189–224.
- Liu, X. and Zhang, X. (2024): “Geopolitical Risk and Currency Returns.” In: *Journal of Banking & Finance* 161. Article 107097.
- Lustig, H., Roussanov, N. and Verdelhan, A. (2011): “Common Risk Factors in Currency Markets.” In: *Review of Financial Studies* 24.11, pp. 3731–3777.
- MacKinlay, A. C. (1997): “Event Studies in Economics and Finance.” In: *Journal of Economic Literature* 35.1, pp. 13–39.
- Newey, W. K. and West, K. D. (1987): “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix.” In: *Econometrica* 55.3, pp. 703–708.
- Obstfeld, M. and Zhou, H. (2023): *The Global Dollar Cycle*. NBER Working Paper 31004. National Bureau of Economic Research.
- Ostry, D., Lloyd, S. P. and Corsetti, G. (2025): “Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations.” In: *Bank of England Staff Working Paper* 1139.
- Ranaldo, A. and Söderlind, P. (2010): “Safe Haven Currencies.” In: *Review of Finance* 14.3, pp. 385–407.
- Salisu, A. A., Olaniran, A. and Tchankam, J.-P. (2022): “Oil Tail Risk and the Tail Risk of the US Dollar Exchange Rates.” In: *Energy Economics* 109. Article 105960.
- Supreme Court of the United States (2026): *Learning Resources, Inc. v. Trump*. Nos. 24–1287 and 25–250, decided 20 February 2026. URL: https://www.supremecourt.gov/opinions/25pdf/24-1287_4gcj.pdf (visited on 08/11/2026).
- U.S. Energy Information Administration (Mar. 2022): *Crude Oil Prices Rise above \$100 per Barrel after Russia’s Further Invasion into Ukraine*. Today in Energy. 4 March 2022. U.S. Energy Information Administration. URL: <https://www.eia.gov/todayinenergy/detail.php?id=55020> (visited on 08/06/2026).
- U.S. Energy Information Administration (Jan. 2026a): *Crude Oil Prices Fell in 2025 amid Oversupply*. Today in Energy. 5 January 2026. U.S. Energy Information

9 References

- Administration. URL: <https://www.eia.gov/todayinenergy/detail.php?id=66944> (visited on 08/06/2026).
- (July 2026c): *Petroleum Markets Responded to Disruptions in the Middle East in the Second Quarter*. Today in Energy. 15 July 2026. U.S. Energy Information Administration. URL: <https://www.eia.gov/todayinenergy/detail.php?id=67865> (visited on 08/06/2026).
- Yilmazkuday, H. (2025): “Geopolitical Risks and Exchange Rates.” In: *Finance Research Letters* 74. Article 106769.

10 Data Sources

- Board of Governors of the Federal Reserve System (2026a): *Foreign Exchange Rates, H.10 Release (21 daily bilateral dollar rates; cross-check panel)*. Daily noon buying rates in New York. Retrieved via FRED, Federal Reserve Bank of St. Louis. URL: <https://www.federalreserve.gov/releases/h10/> (visited on 07/23/2026).
- (2026b): *Market Yields on U.S. Treasury Securities at 10- and 5-Year Constant Maturity; 5-Year TIPS Yield; 5-Year Breakeven Inflation Rate (DGS10, DGS5, DFII5, T5YIE)*. Daily, H.15 release. Retrieved via FRED, Federal Reserve Bank of St. Louis. URL: <https://fred.stlouisfed.org/series/DGS10> (visited on 07/23/2026).
 - (2026c): *Nominal Broad U.S. Dollar Index; Advanced Foreign Economies and Emerging Market Economies Dollar Indexes*. Daily; series DTWEXBGS, DTWEXAFEGS and DTWEXEMEGS. Retrieved via FRED, Federal Reserve Bank of St. Louis. URL: <https://fred.stlouisfed.org/series/DTWEXBGS> (visited on 07/23/2026).
- Caldara, D. and Iacoviello, M. (2026): *Daily Geopolitical Risk (GPR) Index*. Daily. Index of Caldara and Iacoviello (2022). URL: <https://www.matteoiacoviello.com/gpr.htm> (visited on 07/23/2026).
- Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A. and Raffo, A. (2026): *Daily Trade Policy Uncertainty (TPU) Index*. Daily. Index of Caldara et al. (2020); current vintage. URL: <https://www.matteoiacoviello.com/tpu.htm> (visited on 07/23/2026).
- Cboe Global Markets (2026): *Cboe Volatility Index (VIXCLS)*. Daily close. Retrieved via FRED, Federal Reserve Bank of St. Louis. URL: <https://fred.stlouisfed.org/series/VIXCLS> (visited on 07/23/2026).
- LSEG (2026a): *Gold Spot (XAU=); EURO STOXX 50 (.STOXX50E); TTF Natural Gas Front Month (TRNLTTFMc1); Brent Front Month (LCOc1, cross-check)*. Daily. Retrieved via the LSEG Workspace desktop application. URL: <https://www.lseg.com/en/data-analytics/products/workspace> (visited on 07/23/2026).
- (2026b): *J.P. Morgan VXY Global Index of Implied U.S. Dollar Exchange-Rate Volatility (.JPMVXYGL)*. Daily mid quotes, manual price-history export. Retrieved via the LSEG Workspace desktop application. URL: <https://www.lseg.com/en/data-analytics/products/workspace> (visited on 07/23/2026).
 - (2026c): *One-Month Forward Points, same 32-currency panel (RICs <CCY>1M=)*. Daily mid rates. Retrieved via the LSEG Workspace desktop application. URL:

<https://www.lseg.com/en/data-analytics/products/workspace> (visited on 07/23/2026).

LSEG (2026d): *Spot Exchange Rates, End-of-Day Composite Mid Quotes: 32 currencies against the U.S. dollar (RICs EUR= through KWD=)*. Daily mid rates. Retrieved via the LSEG Workspace desktop application. URL: <https://www.lseg.com/en/data-analytics/products/workspace> (visited on 07/23/2026).

S&P Dow Jones Indices (2026): *S&P 500 Index (SP500)*. Daily close. Retrieved via FRED, Federal Reserve Bank of St. Louis. URL: <https://fred.stlouisfed.org/series/SP500> (visited on 07/23/2026).

U.S. Energy Information Administration (2026b): *Crude Oil Prices: Brent—Europe and West Texas Intermediate; Henry Hub Natural Gas Spot Price (DCOILBRETEU, DCOILWTICO, DHHNGSP)*. Daily. Retrieved via FRED, Federal Reserve Bank of St. Louis. URL: <https://fred.stlouisfed.org/series/DCOILBRETEU> (visited on 07/23/2026).

United Nations Statistics Division (2026): *UN Comtrade Database: Annual Trade in Crude Petroleum Oils (HS 2709), 2019–2024*. Annual exports and imports by reporter country, partner World. URL: <https://comtradeplus.un.org> (visited on 08/14/2026).

Appendix

1 Currency Universe and Classification

Table 1 lists the 32 currencies in the panel with their carry-sorted bucket (low/middle/high forward-discount tercile, pegged and managed currencies excluded from the sort) and their oil-trade role. The literature-standard robustness baskets used in Chapter 6 are the yen, Swiss franc and euro (safe) against the Turkish lira, Brazilian real, South African rand, Mexican peso, Colombian peso and Indonesian rupiah (risky).

Table 1: Currency classification

Code	Currency	Carry bucket	Oil role
AUD	Australian dollar	middle	—
BRL	Brazilian real	no 1M forward	exporter
CAD	Canadian dollar	middle	exporter
CHF	Swiss franc	low (safe)	—
CLP	Chilean peso	no 1M forward	—
CNY	Chinese yuan	pegged/managed	—
COP	Colombian peso	no 1M forward	exporter
CZK	Czech koruna	middle	—
DKK	Danish krone	pegged (EUR)	—
EUR	Euro	low (safe)	—
GBP	Pound sterling	middle	—
HKD	Hong Kong dollar	pegged (USD)	—
HUF	Hungarian forint	high (risky)	—
IDR	Indonesian rupiah	high (risky)	—
ILS	Israeli shekel	low (safe)	—
INR	Indian rupee	high (risky)	importer
JPY	Japanese yen	low (safe)	importer
KRW	South Korean won	middle	importer
KWD	Kuwaiti dinar	pegged	exporter
MXN	Mexican peso	high (risky)	exporter
MYR	Malaysian ringgit	middle	—
NOK	Norwegian krone	middle	exporter
NZD	New Zealand dollar	middle	—
PEN	Peruvian sol	no 1M forward	—
PLN	Polish zloty	high (risky)	—
SAR	Saudi riyal	pegged	exporter
SEK	Swedish krona	low (safe)	—
SGD	Singapore dollar	middle	—
THB	Thai baht	low (safe)	importer
TRY	Turkish lira	high (risky)	importer
TWD	Taiwan dollar	low (safe)	importer
ZAR	South African rand	high (risky)	—

Notes. Carry buckets are terciles of the average one-month forward discount (Section 5.1.3). Pegged/managed currencies and the four currencies without a clean one-month forward (BRL, CLP, COP, PEN) are excluded from the carry sort. Oil role follows net crude oil trade over 2019–2024 (Section 5.1.3). In the Comtrade extract one trade direction is unreported for India, Taiwan, Saudi Arabia and Kuwait, Japan’s exports are reported only for 2021–2022 and Mexico’s imports only for 2019. These unreported flows are treated as zero. Reversing a classification would require the missing flow to exceed the absolute observed net trade balance, which ranges from 134 to 1,042 billion dollars over 2019–2024 for these six countries. Türkiye is measured from the mirror side as described in Section 5.1.3 (28.6 billion dollars of imports) and its own crude exports are assumed to be negligible.

2 Robustness

Market model. Alongside the constant-mean specification, the event study computes market-model abnormal returns, using the equal-weighted dollar basket as the market return (Code/03_event_study.py, method `market_model`). Table 2 reports the basket-level results: the safe-haven ordering under both treatment shocks is unchanged relative to the constant-mean results in Chapter 6.

Table 2: Market-model CARs of the currency baskets

	Ukraine		Liberation Day		12-day war		Hormuz	
	(0,5)	(0,20)	(0,5)	(0,20)	(0,5)	(0,20)	(0,5)	(0,20)
Safe currencies	-0.31	-1.73**	+1.31***	+1.17*	-0.13	+0.21	+0.67**	+0.67
Risky currencies	-0.89	-0.21	-0.33	-0.48	-0.26	+0.31	-0.73***	-1.20***
Oil exporters	+1.26***	+2.29***	-0.62	-0.77	+0.34	+0.10	+0.61	+1.23*
Oil importers	+0.51	-1.97	+1.00***	+0.95	-0.49	-0.82	-0.00	-0.02
Carry (risky-safe)	-0.57	+1.52	-1.63***	-1.65	-0.13	+0.10	-1.40***	-1.87**

Notes. Cumulative abnormal returns of the currency baskets in percent. Abnormal returns are residuals from a regression of each basket on the equal-weighted dollar basket estimated over the same $[-140, -21]$ window (Section 5.2). The safe-haven ordering under both treatment shocks is unchanged. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Benchmark and persistence figures. Figure 1 shows the full portfolio response around the external Ukraine benchmark at the identification window. In the main text the Ukraine dollar path enters only through the cross-event differences of Figure 6.4. Figure 2 extends the Liberation Day window to +50 days, plotted from -10 to +50. The tariff drift deepens rather than reverts, the mirror image of the transitory Hormuz premium in Figure 6.5.

Regression robustness. The safe-haven regression of Section 6.6 uses Newey–West (HAC) standard errors, so its inference is robust to the autocorrelation and volatility clustering of daily returns. This is the main reason a simple least-squares interaction is preferred here to a conditional-correlation model. The crisis windows are the same event-day-to-+20 windows used in the event study, so the regression and the event study rest on a common, transparent definition of each episode.

Appendix

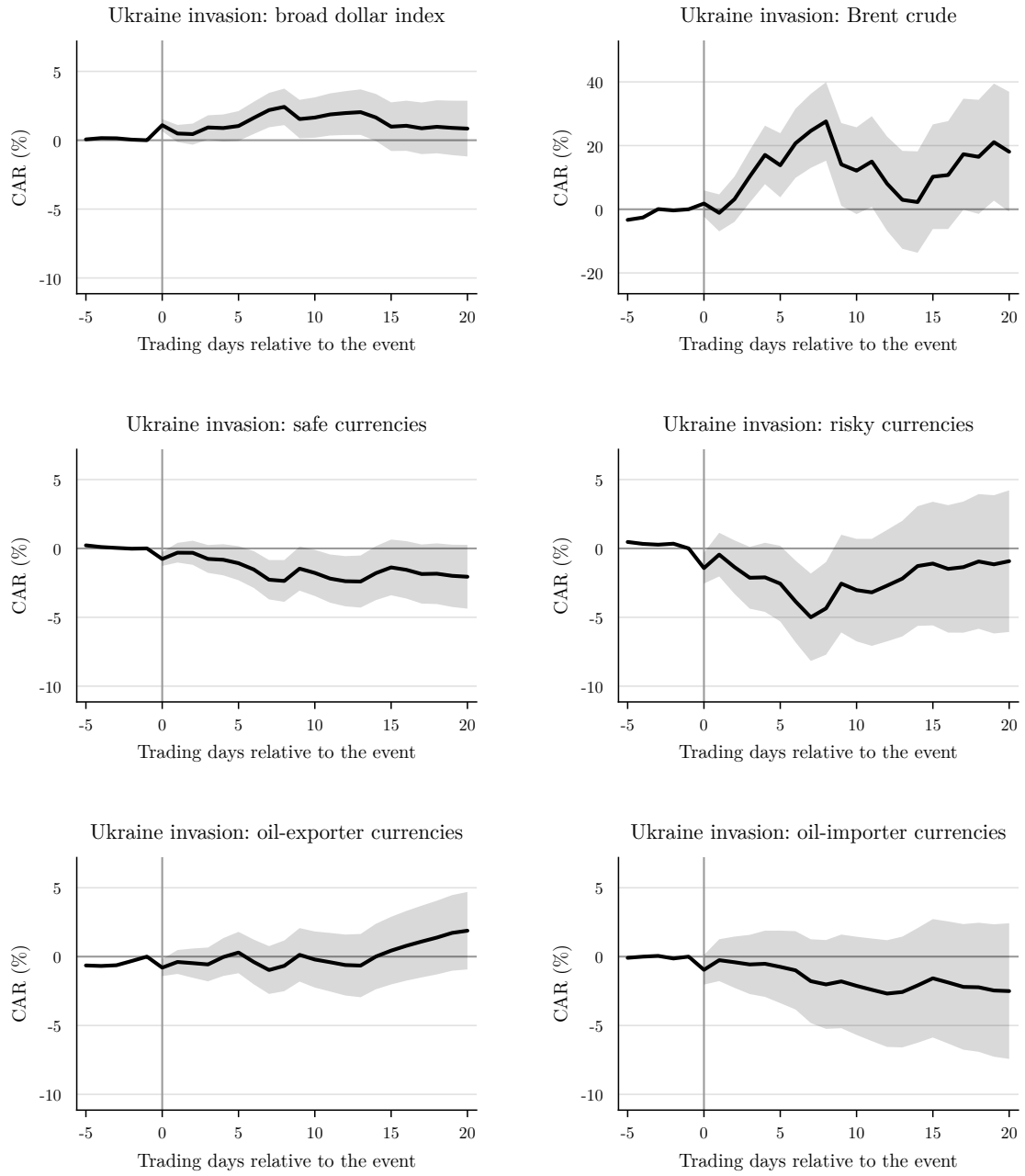


Figure 1: Ukraine invasion (24 February 2022): cumulative abnormal returns over -5 to $+20$ trading days. Construction as in Figure 6.1.

Appendix

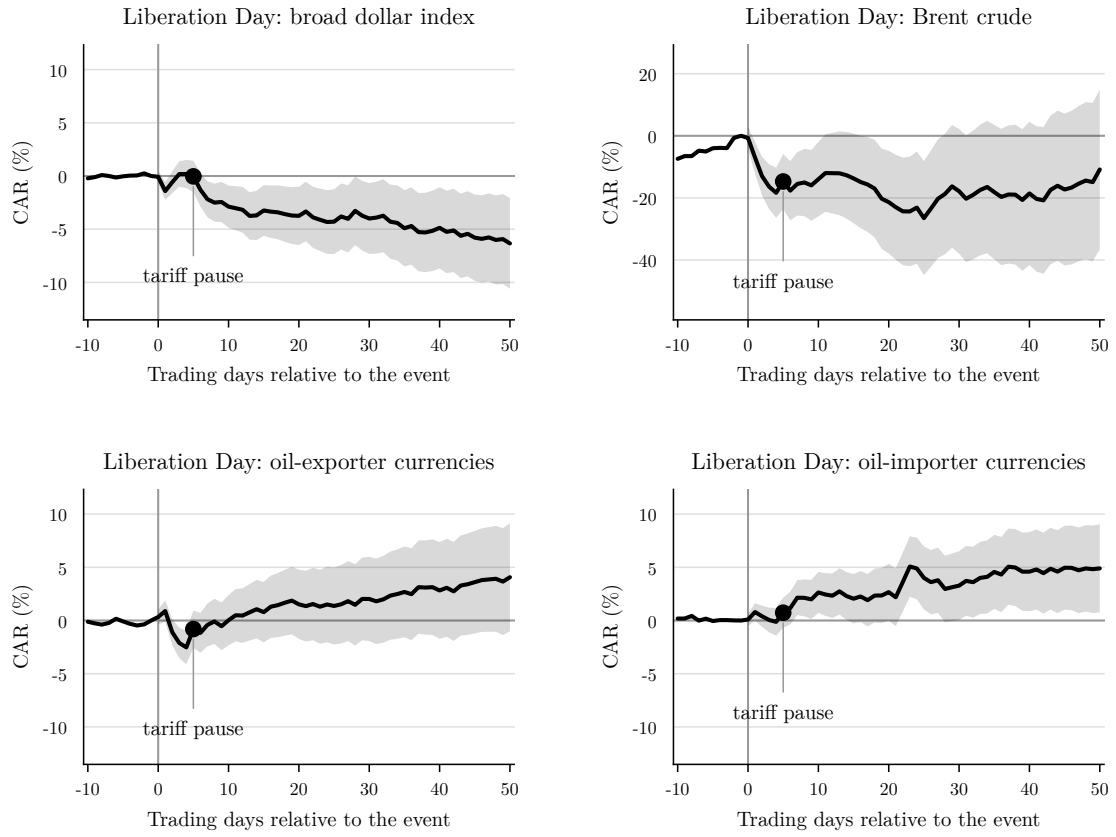


Figure 2: Liberation Day, -10 to $+50$ trading days, estimation window $[-170, -51]$. Construction as in Figure 6.1. The post-announcement dollar decline deepens over the horizon. The 9 April tariff pause (marked) interrupts but does not reverse it.

Affidavit

Ich versichere, dass ich die vorliegende Arbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Ich bin damit einverstanden, dass meine Arbeit zum Zwecke eines Plagiatsabgleichs in elektronischer Form anonymisiert versendet und gespeichert werden kann.

I affirm that this Bachelor thesis was written by myself without any unauthorised third-party support. All used references and resources are clearly indicated. All quotes and citations are properly referenced. This thesis was never presented in the past in the same or similar form to any examination board.

I agree that my thesis may be subject to electronic plagiarism check. For this purpose an anonymous copy may be distributed and uploaded to servers within and outside the University of Mannheim.

Mannheim, September 15, 2026

Signature